

SOME APPLICATIONS

In this section we describe some applications of the theory developed in the book in a variety of different mathematical contexts. These applications are meant to give the reader a flavour of the kind of insight that classifying toposes can offer on ‘concrete’ questions arising in classical mathematics, and by no means exhaust the scope of applicability of the general theory. In fact, the main methodology that we shall use to generate such applications, namely the ‘bridge technique’ described in section 2.2, can, by virtue of its generality, be applied in a great number of different situations.

We shall in particular discuss restrictions of Morita equivalences to quotients of the two theories, give a solution to a problem of Lawvere concerning the boundary operator on subtoposes, establish syntax-semantics ‘bridges’ for quotients of theories of presheaf type, present topos-theoretic interpretations and generalizations of Fraïssé’s theorem in model theory on countably categorical theories and of topological Galois theory, develop a notion of maximal spectrum of a commutative ring with unit and investigate generalized compactness conditions for geometric theories allowing us to identify theories lying in smaller fragments of geometric logic.

10.1 Restrictions of Morita equivalences

It follows as an immediate consequence of the duality of Theorem 3.2.5 that every Morita equivalence between two geometric theories yields, by restriction, Morita equivalences between their quotients. More specifically, we have the following result:

Theorem 10.1.1 *Let $\mathbb{T}_1$ and $\mathbb{T}_2$ be two Morita-equivalent theories. Then for any quotient $\mathbb{S}_1$ of $\mathbb{T}_1$ there exists a unique quotient $\mathbb{S}_2$ of $\mathbb{T}_2$ (up to syntactic equivalence) such that the Morita equivalence between $\mathbb{T}_1$ and $\mathbb{T}_2$ restricts to a Morita equivalence between $\mathbb{S}_1$ and $\mathbb{S}_2$:*

$$\begin{array}{ccc} \mathbb{T}_1\text{-mod}(\mathcal{E}) & \simeq & \mathbb{T}_2\text{-mod}(\mathcal{E}) \\ \uparrow & & \uparrow \\ \mathbb{S}_1\text{-mod}(\mathcal{E}) & \simeq & \mathbb{S}_2\text{-mod}(\mathcal{E}). \end{array}$$

Proof The theorem arises from the following ‘bridge’, whose arches are provided by Theorem 3.2.5:

$$\begin{array}{ccc}
 & \text{Subtoposes of} & \\
 & \mathbf{Sh}(\mathcal{C}_{\mathbb{T}_1}, J_{\mathbb{T}_1}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}_2}, J_{\mathbb{T}_2}) & \\
 \text{Quotients of } \mathbb{T}_1 & \text{---} & \text{Quotients of } \mathbb{T}_2
 \end{array}$$

□

Remark 10.1.2 This theorem would be trivial if the notion of Morita equivalence ‘collapsed’ to bi-interpretability (as defined in section 2.1.3) but, as we have remarked in section 2.2.2, this is not the case. In fact, there are many pairs of theories which are Morita-equivalent without being bi-interpretable; in such situations, the theories cannot be related to each other directly, that is by using a syntactic ‘dictionary’ allowing us to translate formulae written in the language of one theory into formulae written in the language of the other, but Theorem 10.1.1, as well as other results obtained by applying the ‘bridge’ technique in connection with other invariants of classifying toposes, apply, allowing a non-trivial transfer of properties and results across them. For concrete illustrations of this remark, we refer the reader to [30] and [31], which provide natural examples of theories which are Morita-equivalent without being bi-interpretable.

10.2 A solution to the boundary problem for subtoposes

The third of the open problems in topos theory proposed by W. Lawvere in [61] concerns an operation on the lattice of subtoposes of a given topos, introduced in [60] under the name of *boundary* of a subtopos. The problem asks to describe the effect of the boundary operator on the classes of models classified by these subtoposes, both syntactically and semantically.

Recall from [60] that the boundary of an element a in a co-Heyting algebra C is given by the meet between a and the smallest element c of C such that $a \vee c = 1$. The canonical example of a co-Heyting algebra is the lattice of closed subsets of a topological space; in this case, the notion of boundary specializes to the ordinary topological one (that is, the boundary of a closed subset C is the set of elements of C which do not belong to its interior).

Thanks to the explicit description of the Heyting algebra operations on quotients of a given geometric theory obtained in sections 4.1.2 and 4.1.3, and to the description of the Heyting operations on Grothendieck topologies established in section 4.1.1, we can give an answer to both of the above-mentioned questions.

Indeed, the operation of boundary on subtoposes of the classifying topos of a geometric theory $\mathbb{T}$ over a signature Σ corresponds, via the duality of Theorem 3.2.5, to the operation $\mathbb{T}' \rightarrow \mathbb{T}' \vee \neg \mathbb{T}'$ in the lattice $\mathfrak{H}_{\Sigma}^{\mathbb{T}}$. The explicit description of the lattice and Heyting operations on $\mathfrak{H}_{\Sigma}^{\mathbb{T}}$ obtained in sections 4.1.2 and 4.1.3

thus provides an explicit axiomatization of the quotient $\partial\mathbb{T}'$ of $\mathbb{T}$ corresponding to the boundary of the subtopos $\mathbf{Set}[\mathbb{T}']$ of $\mathbf{Set}[\mathbb{T}]$ associated with a quotient $\mathbb{T}'$ of $\mathbb{T}$ via Theorem 3.2.5, as follows: $\partial\mathbb{T}'$ is obtained from $\mathbb{T}'$ by adding all the geometric sequents $(\psi \vdash_{\vec{y}} \psi')$ over Σ with the property that $(\psi' \vdash_{\vec{y}} \psi)$ is provable in $\mathbb{T}$ and for any $\mathbb{T}$ -provably functional geometric formula $\theta(\vec{x}, \vec{y})$ from a geometric formula-in-context $\{\vec{x}. \phi\}$ to $\{\vec{y}. \psi\}$ and any geometric formula χ in the context $\vec{x}$ such that $(\chi \vdash_{\vec{x}} \phi)$ is provable in $\mathbb{T}$, the conjunction of the facts

- (i) $(\phi \vdash_{\vec{x}} \chi)$ is provable in $\mathbb{T}'$
- (ii) $((\exists \vec{y})(\theta(\vec{x}, \vec{y}) \wedge \psi'(\vec{y})) \vdash_{\vec{x}} \chi)$ is provable in $\mathbb{T}$

implies that $(\phi \vdash_{\vec{x}} \chi)$ is provable in $\mathbb{T}$.

Concerning the problem of getting a semantic description of the models of the quotient $\mathbb{T}'$ of a theory of presheaf type $\mathbb{T}$ corresponding to the boundary of a given subtopos $\mathcal{F}$ of the classifying topos $\mathbf{Set}[\mathbb{T}] \simeq [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ of $\mathbb{T}$, we observe that if $\mathcal{F}$ corresponds to a Grothendieck topology J on the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ as in section 8.1.2 then the models of $\mathbb{T}'$ are precisely the $(J \vee \neg J)$ -homogeneous $\mathbb{T}$ -models (in the sense of Definition 8.1.1).

Of course, these are abstract solutions which work uniformly for any geometric theory (resp. for any theory of presheaf type) $\mathbb{T}$. In practice one is often interested in finding more ‘economical’ axiomatizations which exploit the specific features of the theories under consideration; in this respect, the duality between quotients and Grothendieck topologies established in Chapter 8 and the explicit formulae for the lattice operations on Grothendieck topologies obtained in section 4.1.1 are clearly relevant (cf. for instance section 8.1.5).

10.3 Syntax-semantics ‘bridges’

In this section we establish several results on theories of presheaf type and their quotients by transferring certain invariant properties of (sub)toposes across two different representations of a given classifying topos, according to the technique ‘toposes as bridges’ described in section 2.2.

We have seen in section 6.1.2 that if $\mathbb{T}$ is a theory of presheaf type then, by choosing a canonical Morita equivalence for $\mathbb{T}$ (in the sense of section 6.1.1), we have an equivalence of classifying toposes

$$[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

sending the universal model $N_{\mathbb{T}}$ of $\mathbb{T}$ in the topos $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ (as in Theorem 6.1.1) into the universal model $U_{\mathbb{T}}$ of $\mathbb{T}$ in $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ (as in Remark 2.1.9). In particular, if $M \in \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ is a $\mathbb{T}$ -model presented by a formula $\phi(\vec{x})$ over Σ then, denoting by

$$y : \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}} \rightarrow [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$$

the Yoneda embedding, we have that $y(M)$ is isomorphic to $[[\vec{x}. \phi]]_{N_{\mathbb{T}}}$ and hence corresponds, under the equivalence

$$[\mathbf{f.p.T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}),$$

to the sheaf $y_{\mathbb{T}}(\{\vec{x} \cdot \phi\}) = [[\vec{x} \cdot \phi]]_{U_{\mathbb{T}}}$, where $y_{\mathbb{T}} : \mathcal{C}_{\mathbb{T}} \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ is the Yoneda embedding.

Now, if $\mathbb{T}'$ is a quotient of $\mathbb{T}$ then the subtopos of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ corresponding to it via Theorem 3.2.5 transfers via the equivalence

$$[\mathbf{f.p.T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

to a subtopos

$$\mathbf{Sh}(\mathbf{f.p.T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}, J) \hookrightarrow [\mathbf{f.p.T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$$

of $[\mathbf{f.p.T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$, where J is the induced $\mathbb{T}$ -topology of $\mathbb{T}'$ (as defined in section 8.1.2). This gives rise to an equivalence

$$\mathbf{Sh}(\mathbf{f.p.T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}, J) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'})$$

of classifying toposes of $\mathbb{T}'$ which sends $l_J^{\mathbf{f.p.T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}}(M)$ to $y_{\mathbb{T}'}(\{\vec{x} \cdot \phi\}) = [[\vec{x} \cdot \phi]]_{U_{\mathbb{T}'}}$, where $l_J^{\mathbf{f.p.T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}}$ is the composite of y with the associated sheaf functor

$$[\mathbf{f.p.T}\text{-mod}(\mathbf{Set}), \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathbf{f.p.T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}, J)$$

and $y_{\mathbb{T}'} : \mathcal{C}_{\mathbb{T}'} \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'})$ is the Yoneda embedding.

We shall establish links between ‘geometrical’ properties of J and syntactic properties of $\mathbb{T}'$ by transferring suitable invariant properties of toposes or of objects of toposes across the equivalence

$$\mathbf{Sh}(\mathbf{f.p.T}\text{-mod}(\mathbf{Set})^{\mathrm{op}}, J) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}).$$

In order to perform these ‘translations’ of properties of one site into properties of the other across this equivalence, we shall need characterizations relating properties of (objects of) sites and properties of (objects of) the corresponding toposes, which will play the role of the ‘arches’ of our bridges (in the sense of section 2.2.3).

As ‘global’ invariant properties of toposes we shall use the notions of atomic topos, locally connected topos and the invariant property of a topos to be equivalent to a presheaf topos, while as ‘local’ invariant properties of objects of toposes we shall use the notion of atom, of indecomposable object and of irreducible object.

Before proceeding further, let us recall the precise meaning of the above-mentioned terms. An *atom* of a topos is a non-zero object which does not have any proper subobjects. An *indecomposable object* of a Grothendieck topos is an object which does not admit any non-trivial coproduct decomposition. An *irreducible object* of a Grothendieck topos is an object such that every epimorphic family to it contains a split epimorphism. A Grothendieck topos is said

to be *atomic* if all its subobject lattices are complete atomic Boolean algebras, equivalently if it has a separating set of atoms. A Grothendieck topos is *locally connected* (resp. equivalent to a presheaf topos) if and only if it has a separating set of indecomposable (resp. irreducible) objects. For a proof of these characterizations we refer the reader to [20].

These invariants admit natural well-known site characterizations of the kind ‘if a site $(\mathcal{C}, J)$ satisfies a certain property then the topos $\mathbf{Sh}(\mathcal{C}, J)$ satisfies the given invariant’ (cf. section C3 of [55]), which can be combined with criteria going in the other direction of the form ‘if the classifying topos of a geometric theory $\mathbb{T}$ satisfies the given invariant then the theory $\mathbb{T}$ satisfies a certain property’ to form ‘bridges’ which allow us to deduce properties of a quotient $\mathbb{T}'$ from properties of the associated Grothendieck topology J :

$$\begin{array}{ccc} & \mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) & \\ \nearrow \text{---} & & \searrow \text{---} \\ (\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J) & & \mathbb{T}' \end{array}$$

In order to state the following result, obtained by applying this methodology, it is convenient to introduce some terminology.

Definition 10.3.1 Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\phi(\vec{x})$ a geometric formula-in-context over Σ .

- (a) We say that $\phi(\vec{x})$ is $\mathbb{T}$ -*complete* if the sequent $(\phi \vdash_{\vec{x}} \perp)$ is not provable in $\mathbb{T}$, and for every geometric formula χ in the same context either $(\chi \wedge \phi \vdash_{\vec{x}} \perp)$ or $(\phi \vdash_{\vec{x}} \chi)$ is provable in $\mathbb{T}$ (cf. p. 928 of [55]).
- (b) We say that $\phi(\vec{x})$ is $\mathbb{T}$ -*indecomposable* if for any family $\{\psi_i(\vec{x}) \mid i \in I\}$ of geometric formulae in the same context such that, for each i , ψ_i $\mathbb{T}$ -provably implies ϕ and, for any distinct $i, j \in I$, $(\psi_i \wedge \psi_j \vdash_{\vec{x}} \perp)$ is provable in $\mathbb{T}$, we have that if $(\phi \vdash_{\vec{x}} \bigvee_{i \in I} \psi_i)$ is provable in $\mathbb{T}$ then $(\phi \vdash_{\vec{x}} \psi_i)$ is provable in $\mathbb{T}$ for some $i \in I$ (cf. Definition 3.1(b) [20]).
- (c) We say that $\phi(\vec{x})$ is $\mathbb{T}$ -*irreducible* if for any family $\{\theta_i \mid i \in I\}$ of $\mathbb{T}$ -provably functional geometric formulae $\{\vec{x}_i, \vec{x}. \theta_i\}$ from $\{\vec{x}_i. \phi_i\}$ to $\{\vec{x}. \phi\}$ such that $(\phi \vdash_{\vec{x}} \bigvee_{i \in I} (\exists \vec{x}_i' \theta_i))$ is provable in $\mathbb{T}$, there exist $i \in I$ and a $\mathbb{T}$ -provably functional geometric formula $\{\vec{x}, \vec{x}_i'. \theta'\}$ from $\{\vec{x}. \phi\}$ to $\{\vec{x}_i'. \phi_i\}$ such that the bisequent $(\phi(\vec{x}) \wedge \vec{x} = \vec{x}' \dashv\vdash_{\vec{x}, \vec{x}'} (\exists \vec{x}_i') (\theta'(\vec{x}, \vec{x}_i') \wedge \theta_i(\vec{x}_i', \vec{x}')))$ is provable in $\mathbb{T}$ (cf. Definition 6.1.9).

As shown by the following lemma, the above-mentioned notions arise precisely by expressing in terms of the syntactic site $(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ the property of the image of a geometric formula $\{\vec{x}. \phi\}$ under the Yoneda embedding $y_{\mathbb{T}} : \mathcal{C}_{\mathbb{T}} \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ to be an atom (resp. an indecomposable object, an irreducible object) of the topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$.

Lemma 10.3.2 (cf. Lemma 3.2 [20]). *Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\phi(\vec{x})$ a geometric formula-in-context over Σ . Then*

- (i) *$\phi(\vec{x})$ is $\mathbb{T}$ -complete if and only if $y_{\mathbb{T}}(\{\vec{x}.\phi\})$ is an atom of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$.*
- (ii) *$\phi(\vec{x})$ is $\mathbb{T}$ -indecomposable if and only if $y_{\mathbb{T}}(\{\vec{x}.\phi\})$ is an indecomposable object of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$.*
- (iii) *$\phi(\vec{x})$ is $\mathbb{T}$ -irreducible if and only if $y_{\mathbb{T}}(\{\vec{x}.\phi\})$ is an irreducible object of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$.*

The following site characterizations will provide the other ‘arches’ of our ‘bridges’:

Proposition 10.3.3 (cf. Proposition 2.5 [20]). *Let $(\mathcal{C}, J)$ be a small site and $l_J^{\mathcal{C}} : \mathcal{C} \rightarrow \mathbf{Sh}(\mathcal{C}, J)$ the composite of the Yoneda embedding $\mathcal{C} \hookrightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ with the associated sheaf functor.*

- (i) *If J is trivial then, for any $c \in \mathcal{C}$, $l_J^{\mathcal{C}}(c)$ is an irreducible object of $\mathbf{Sh}(\mathcal{C}, J)$.*
- (ii) *If $(\mathcal{C}, J)$ is locally connected (in the sense that every J -covering sieve S on an object c of $\mathcal{C}$ is connected as a full subcategory of $\mathcal{C}/c$, cf. p. 657 of [55]) then, for each $c \in \mathcal{C}$, $l_J^{\mathcal{C}}(c)$ is an indecomposable object of $\mathbf{Sh}(\mathcal{C}, J)$; in particular, the topos $\mathbf{Sh}(\mathcal{C}, J)$ is locally connected.*
- (iii) *If $(\mathcal{C}, J)$ is atomic (in the sense that $\mathcal{C}$ satisfies the right Ore condition and J is the atomic topology on it) then, for each $c \in \mathcal{C}$, $l_J^{\mathcal{C}}(c)$ is an atom of $\mathbf{Sh}(\mathcal{C}, J)$; in particular, the topos $\mathbf{Sh}(\mathcal{C}, J)$ is atomic.*

We are now ready to state the main theorem of this section.

Theorem 10.3.4 (cf. [20]). *Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ and $\mathbb{T}'$ a quotient of $\mathbb{T}$ with induced $\mathbb{T}$ -topology J on $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ (as in section 8.1.2).*

- (i) *If J is trivial then $\mathbb{T}' (= \mathbb{T})$ satisfies the property that for any geometric formula $\{\vec{y}.\psi\}$ over Σ there exist $\mathbb{T}$ -irreducible formulae-in-context $\{\vec{x}_i.\phi_i\}$ (for $i \in I$) and $\mathbb{T}$ -provably functional geometric formulae $\{\vec{x}_i.\vec{y}.\theta_i\}$ from $\{\vec{x}_i.\phi_i\}$ to $\{\vec{y}.\psi\}$ such that $(\psi \vdash_{\vec{y}} \bigvee_{i \in I} (\exists \vec{x}_i)\theta_i)$ is provable in $\mathbb{T}'$ (cf. Corollary 6.1.10).
Moreover, any formula which presents a $\mathbb{T}$ -model is $\mathbb{T}$ -irreducible (cf. Theorem 6.1.13).*
- (ii) *If the site $(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J)$ is locally connected then $\mathbb{T}'$ satisfies the property that every geometric formula-in-context over Σ is $\mathbb{T}'$ -provably equivalent to a disjunction of $\mathbb{T}'$ -indecomposable geometric formulae in the same context which are $\mathbb{T}'$ -provably disjoint from each other.
Moreover, every formula which presents a $\mathbb{T}$ -model is $\mathbb{T}'$ -indecomposable.*
- (iii) *If the site $(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J)$ is atomic then $\mathbb{T}'$ satisfies the property that every geometric formula-in-context over Σ is $\mathbb{T}'$ -provably equivalent to a disjunction of $\mathbb{T}'$ -complete geometric formulae in the same context.
Moreover, every formula which presents a $\mathbb{T}$ -model is $\mathbb{T}'$ -complete.*

Proof

(i) The first assertion about $\mathbb{T}$ follows from Corollary 6.1.10. Given a geometric formula $\phi(\vec{x})$ over Σ presenting a $\mathbb{T}$ -model M , by Lemma 10.3.2(iii) $\phi(\vec{x})$ is $\mathbb{T}$ -irreducible if and only if $y_{\mathbb{T}}(\{\vec{x}.\phi\})$ is an irreducible object of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$. But this is equivalent to saying that $y(M)$ is irreducible in $[\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$ (cf. the discussion preceding Definition 10.3.1), and this is true since all representables in a presheaf topos are irreducible objects of it (cf. Proposition 10.3.3).

(ii) and (iii) Given a geometric formula $\phi(\vec{x})$ over Σ presenting a $\mathbb{T}$ -model M , by Lemma 10.3.2(ii) (resp. Lemma 10.3.2(i)) $\phi(\vec{x})$ is $\mathbb{T}'$ -indecomposable (resp. $\mathbb{T}'$ -complete) if and only if $y_{\mathbb{T}'}(\{\vec{x}.\phi\})$ is an indecomposable object (resp. an atom) of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'})$; but this means that $l_J^{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}}(M)$ is an indecomposable object (resp. an atom) of $\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J)$, and this is true by Proposition 10.3.3. The above-mentioned characterizations of the invariant property of a Grothendieck topos to be locally connected (resp. atomic) thus yield our thesis. $\square$

Remark 10.3.5 As shown in [21], one can easily introduce new ‘geometric’ invariant properties of toposes other than the ones considered above, defined for instance by requiring the existence of separating sets of objects satisfying particular properties, and establish site characterizations for them yielding ‘bridges’ such as the ones in the proof of Theorem 10.3.4. In other words, Theorem 10.3.4 provides just a few examples of insights relating the syntactic properties of a quotient $\mathbb{T}'$ of a theory of presheaf type $\mathbb{T}$ and the ‘geometric’ properties of the associated Grothendieck topology J which are obtainable by applying the ‘bridge’ technique.

10.4 Topos-theoretic Fraïssé theorem

In this section, which is based on [23], we present a topos-theoretic interpretation and wide generalization of the well-known result (Theorem 7.4.1(a) [49]) providing the link between Fraïssé’s construction and countably categorical theories. The key concepts involved in Fraïssé’s construction (i.e. amalgamation and joint embedding properties, homogeneous structures and atomicity and completeness of the theory axiomatizing them) will be seen to naturally arise from the expression of topos-theoretic invariants on a classifying topos in terms of different sites of definition for it, and the connections between them to result from ‘bridges’ (in the sense of section 2.2) involving these different representations.

The context in which we shall formulate our topos-theoretic interpretation of Fraïssé’s theorem is that of theories of presheaf type, which we comprehensively studied in Chapter 6. Recall that this class of theories is very extensive and includes all the cartesian theories as well as many other interesting theories arising in different fields of mathematics (cf. also Chapter 9).

The following notions, which will play a central role in our analysis, are natural categorical generalizations of the concepts involved in the classical Fraïssé construction.

Definition 10.4.1 A category $\mathcal{C}$ is said to satisfy the *amalgamation property* (AP) if for any objects $a, b, c \in \mathcal{C}$ and arrows $f : a \rightarrow b, g : a \rightarrow c$ in $\mathcal{C}$ there exist an object $d \in \mathcal{C}$ and arrows $f' : b \rightarrow d, g' : c \rightarrow d$ in $\mathcal{C}$ such that $f' \circ f = g' \circ g$:

$$\begin{array}{ccc} a & \xrightarrow{f} & b \\ g \downarrow & & \downarrow f' \\ c & \xrightarrow{g'} & d. \end{array}$$

The amalgamation property on a category is also called the left Ore condition, and its dual the right Ore condition. Notice that if $\mathcal{C}$ satisfies AP then we can equip $\mathcal{C}^{\text{op}}$ with the atomic topology (that is, the Grothendieck topology whose covering sieves are exactly the non-empty ones). This point will be a fundamental ingredient of our topos-theoretic interpretation of Fraïssé's theorem.

Definition 10.4.2 A category $\mathcal{C}$ is said to satisfy the *joint embedding property* (JEP) if for every pair of objects $a, b \in \mathcal{C}$ there exist an object $c \in \mathcal{C}$ and arrows $f : a \rightarrow c, g : b \rightarrow c$ in $\mathcal{C}$:

$$\begin{array}{c} a \\ | \\ b \end{array} \xrightarrow{g} \begin{array}{c} a \\ | \\ c \end{array} \begin{array}{c} f \\ | \\ g \end{array}$$

Notice that if $\mathcal{C}$ has a weakly initial object then AP on $\mathcal{C}$ implies JEP on $\mathcal{C}$; however, in general the two notions are distinct from each other.

Definition 10.4.3 Let $\mathcal{C} \hookrightarrow \mathcal{D}$ be an embedding of categories.

- (a) An object $u \in \mathcal{D}$ is said to be $\mathcal{C}$ -*homogeneous* if for any objects $a, b \in \mathcal{C}$ and arrows $j : a \rightarrow b$ in $\mathcal{C}$ and $\chi : a \rightarrow u$ in $\mathcal{D}$ there exists an arrow $\tilde{\chi} : b \rightarrow u$ in $\mathcal{D}$ such that $\tilde{\chi} \circ j = \chi$:

$$\begin{array}{ccc} a & \xrightarrow{\chi} & u \\ j \downarrow & \nearrow \tilde{\chi} & \\ b & & \end{array}$$

- (b) An object $u \in \mathcal{D}$ is said to be $\mathcal{C}$ -*ultrahomogeneous* if for any objects $a, b \in \mathcal{C}$ and arrows $j : a \rightarrow b$ in $\mathcal{C}$ and $\chi_1 : a \rightarrow u, \chi_2 : b \rightarrow u$ in $\mathcal{D}$ there exists an isomorphism $\tilde{j} : u \rightarrow u$ in $\mathcal{D}$ such that $\tilde{j} \circ \chi_1 = \chi_2 \circ j$:

$$\begin{array}{ccc} a & \xrightarrow{\chi_1} & u \\ j \downarrow & & \downarrow \tilde{j} \\ b & \xrightarrow{\chi_2} & u. \end{array}$$

- (c) An object $u \in \mathcal{D}$ is said to be $\mathcal{C}$ -universal if for every $a \in \mathcal{C}$ there exists an arrow $\chi : a \rightarrow u$ in $\mathcal{D}$:

$$a - \frac{\chi}{} \succ u.$$

Remarks 10.4.4 (a) Any $\mathcal{C}$ -ultrahomogeneous and $\mathcal{C}$ -universal object is $\mathcal{C}$ -homogeneous.

- (b) In verifying that an object u in $\mathcal{D}$ is $\mathcal{C}$ -ultrahomogeneous one can clearly suppose, without loss of generality, that the arrow j in Definition 10.4.3(b) is an identity.

Let $\mathbb{T}$ be a theory of presheaf type over a signature Σ . If the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ of its finitely presentable models satisfies the amalgamation property then we can put on $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ the atomic topology J_{at} , obtaining a subtopos

$$\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J) \hookrightarrow [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}), \mathbf{Set}]$$

of the classifying topos of $\mathbb{T}$ which, once a canonical Morita equivalence for $\mathbb{T}$ (in the sense of section 6.1.1) is chosen, corresponds by the duality of Theorem 3.2.5 to a unique quotient $\mathbb{T}'$ of $\mathbb{T}$ classified by it. This quotient can be characterized, by Corollary 8.1.14, as the theory over Σ obtained from $\mathbb{T}$ by adding all the sequents of the form $(\psi \vdash_{\vec{y}} (\exists \vec{x})\theta(\vec{x}, \vec{y}))$, where $\phi(\vec{x})$ and $\psi(\vec{y})$ are formulae which present a $\mathbb{T}$ -model and $\theta(\vec{x}, \vec{y})$ is a $\mathbb{T}$ -provably functional formula over Σ from $\{\vec{x}. \phi\}$ to $\{\vec{y}. \psi\}$.

Since $\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}})$ classifies the theory $\mathbb{T}'$, by the syntactic method for constructing classifying toposes (cf. Theorem 2.1.8), our canonical Morita equivalence for $\mathbb{T}$ induces an equivalence

$$\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}),$$

which can be used for building ‘bridges’ between the two sites by considering appropriate topos-theoretic invariants on the classifying topos of $\mathbb{T}'$:

$$\begin{array}{ccc} & \mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) & \\ \text{---} \swarrow & & \searrow \text{---} \\ (\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) & & (\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) \end{array}$$

First, we observe that, by Theorem 8.1.3 and Remark 8.1.2, the theory $\mathbb{T}'$ axiomatizes the homogeneous $\mathbb{T}$ -models (in the sense of Remark 8.1.2) in every Grothendieck topos $\mathcal{E}$:

$$\begin{array}{ccc} & \text{Geometric morphism from } \mathcal{E} \text{ to} & \\ & \mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) & \\ \text{---} \swarrow & & \searrow \text{---} \\ (\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) & & (\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) \\ \text{homogeneous } \mathbb{T}\text{-model in } \mathcal{E} & & \mathbb{T}'\text{-model in } \mathcal{E} \end{array}$$

In particular, taking $\mathcal{E} = \mathbf{Set}$, we obtain that the set-based $\mathbb{T}'$ -models are the set-based models M of $\mathbb{T}$ such that for any arrow $y : c \rightarrow M$ in $\mathbb{T}\text{-mod}(\mathbf{Set})$ and

any arrow $f : c \rightarrow d$ in $\mathbf{f.p.T-mod}(\mathbf{Set})$ there exists an arrow u in $\mathbf{T-mod}(\mathbf{Set})$ such that $u \circ f = y$:

$$\begin{array}{ccc} c & \xrightarrow{y} & M \\ f \downarrow & \nearrow u & \\ d & & \end{array}$$

For this reason, we shall call $\mathbf{T}'$ the ‘theory of homogeneous $\mathbf{T}$ -models’. Notice that a set-based $\mathbf{T}$ -model M is homogeneous if and only if it is $\mathbf{f.p.T-mod}(\mathbf{Set})$ -homogeneous as an object of the category $\mathbf{T-mod}(\mathbf{Set})$ in the sense of Definition 10.4.3.

Next, let us consider the invariant property of the classifying topos of $\mathbf{T}'$ to be atomic. The fact that $(\mathbf{f.p.T-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}})$ is an atomic site implies, by a well-known site characterization (cf. also Proposition 10.3.3), that the topos $\mathbf{Sh}(\mathbf{f.p.T-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}})$ is atomic. By looking at this invariant from the point of view of the other site $(\mathcal{C}_{\mathbf{T}'}, J_{\mathbf{T}'})$ we obtain, as an application of the classical characterization of the geometric theories whose classifying topos is atomic (cf. Proposition D3.4.13 [50]), that the theory $\mathbf{T}'$ is atomic in a logical sense (cf. Theorem 10.3.4(iii)), i.e. that every geometric formula-in-context over its signature is $\mathbf{T}'$ -provably equivalent to a disjunction of $\mathbf{T}'$ -complete formulae in the same context (cf. Definition 10.3.1(a)):

$$\begin{array}{ccc} & \xrightarrow{\text{Atomic topos}} & \mathbf{Sh}(\mathbf{f.p.T-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbf{T}'}, J_{\mathbf{T}'}) \\ \text{---} \nearrow & & \nwarrow \text{---} \\ (\mathbf{f.p.T-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) & & (\mathcal{C}_{\mathbf{T}'}, J_{\mathbf{T}'}) \\ \text{AP on } \mathbf{f.p.T-mod}(\mathbf{Set}) & & \text{Atomicity of } \mathbf{T}' \end{array}$$

Moreover, recall from Theorem 10.3.4(iii) that every geometric formula over Σ which presents a $\mathbf{T}$ -model is $\mathbf{T}'$ -complete. Indeed, denoting by

$$y_{\mathbf{T}'} : \mathcal{C}_{\mathbf{T}'} \rightarrow \mathbf{Sh}(\mathcal{C}_{\mathbf{T}'}, J_{\mathbf{T}'})$$

the Yoneda embedding and by

$$l_{J_{\text{at}}} : \mathbf{f.p.T-mod}(\mathbf{Set})^{\text{op}} \rightarrow \mathbf{Sh}((\mathbf{f.p.T-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}))$$

the composite of the Yoneda embedding

$$\mathbf{f.p.T-mod}(\mathbf{Set})^{\text{op}} \rightarrow [\mathbf{f.p.T-mod}(\mathbf{Set}), \mathbf{Set}]$$

with the associated sheaf functor

$$[\mathbf{f.p.T-mod}(\mathbf{Set}), \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathbf{f.p.T-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}),$$

we have that for any finitely presentable $\mathbf{T}$ -model M the objects $l_{J_{\text{at}}}(M)$ and $y_{\mathbf{T}'}(\{\vec{x}. \phi\})$, where $\phi(\vec{x})$ is ‘the’ formula presenting M , correspond to each other

under the Morita equivalence for $\mathbb{T}'$ considered above; now, the fact that J_{at} is the atomic topology implies that $l_{J_{\text{at}}}(M)$ is an atom (cf. Proposition 10.3.3), while the property of $y_{\mathbb{T}'}(\{\vec{x}. \phi\})$ to be an atom is precisely equivalent to the condition of $\mathbb{T}'$ -completeness of the formula $\phi(\vec{x})$ (cf. Lemma 10.3.2(i)). In general, it is not true that all the atoms of the classifying topos of $\mathbb{T}'$ have this form (for example, for any discrete group G , regarded as a one-object category, the topos $[G^{\text{op}}, \mathbf{Set}]$ is atomic and its atoms can be identified with the transitive G -sets, while there is only one representable, namely the functor sending the unique object of G to the underlying set of G), but the fact that in an atomic topos every atom can be covered by a representable ensures that if an object of the form $y_{\mathbb{T}'}(\{\vec{y}. \psi\})$ is an atom, then there exists a $\mathbb{T}'$ -provably functional formula $\theta(\vec{x}, \vec{y})$ from a formula $\{\vec{x}. \phi\}$ presenting a $\mathbb{T}$ -model to $\{\vec{y}. \psi\}$ with the property that the sequent $(\psi \vdash_{\vec{y}} (\exists \vec{x})\theta(\vec{x}, \vec{y}))$ is provable in $\mathbb{T}'$. We shall characterize the subcanonical atomic sites such that every atom of the corresponding topos comes from the site (up to isomorphism) in section 10.5 (cf. Theorem 10.5.6).

Let us now consider the invariant property of a topos to be *two-valued* (in the sense that its only two subterminal objects are the zero and the identity one, and they are distinct from each other) in connection with the above Morita equivalence for $\mathbb{T}'$. By using the characterization of the subterminal objects of a topos $\mathbf{Sh}(\mathcal{C}, J)$ in terms of J -ideals on $\mathcal{C}$ provided by Remark 1.1.25, it is easy to see that for any atomic site $(\mathcal{C}, J_{\text{at}})$ such that $\mathcal{C}$ is non-empty, the topos $(\mathcal{C}, J_{\text{at}})$ is two-valued if and only if $\mathcal{C}^{\text{op}}$ satisfies the joint embedding property (cf. Theorem 3.6 and Lemma 3.7 [23]). On the other hand, applying this characterization in the case of the classifying topos of a geometric theory $\mathbb{S}$, represented as the topos of sheaves on its geometric syntactic site, yields the following criterion: the topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{S}}, J_{\mathbb{S}})$ is two valued if and only if the theory $\mathbb{S}$ is *complete*, in the sense that every geometric sentence over the signature of $\mathbb{S}$ is either provably false or provably true, but not both – cf. Definition 4.2.21 (notice that the subterminal objects of a topos $\mathbf{Sh}(\mathcal{C}_{\mathbb{S}}, J_{\mathbb{S}})$ correspond precisely to the $\mathbb{S}$ -provable equivalence classes of geometric sentences over the signature of $\mathbb{S}$). Notice that if a theory is complete then all its set-based models satisfy the same geometric sentences; the converse holds if the theory has enough set-based models. This notion of completeness for geometric theories notably agrees with the classical model-theoretic one for (finitary) first-order theories when applied to atomic theories (cf. Remark 10.4.7 below).

By applying these criteria in the context of the Morita equivalence

$$\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}),$$

we obtain the following logical equivalence: the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$, if it is non-empty, satisfies JEP if and only if the theory $\mathbb{T}'$ is complete:

$$\begin{array}{ccc}
 & \text{Two-valued topos} & \\
 & \mathbf{Sh}(\mathbf{f.p.T-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) & \\
 \text{(f.p.T-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) & \text{---} & (\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) \\
 \text{JEP on f.p.T-mod}(\mathbf{Set}) & & \text{Completeness of } \mathbb{T}'
 \end{array}$$

Finally, recall from [18] (cf. Theorem 3.15 therein) that, assuming the axiom of countable choice, every geometric theory which is atomic and complete is countably categorical, that is any two of its set-based countable models are isomorphic.

Summarizing, we have the following

Theorem 10.4.5 (Theorem 3.8 [23]). *Let $\mathbb{T}$ be a theory of presheaf type such that the category $\mathbf{f.p.T-mod}(\mathbf{Set})$ is non-empty and satisfies the amalgamation and joint embedding properties. Then (any theory Morita-equivalent to) the theory $\mathbb{T}'$ of homogeneous $\mathbb{T}$ -models is complete and atomic; in particular, assuming the axiom of countable choice, any two countable homogeneous $\mathbb{T}$ -models in $\mathbf{Set}$ are isomorphic.*

Moreover, every geometric formula which presents a $\mathbb{T}$ -model is $\mathbb{T}'$ -complete. $\square$

Remark 10.4.6 If the category $\mathbf{f.p.T-mod}(\mathbf{Set})$ satisfies AP then each of its connected components satisfies AP as well as JEP. The toposes of sheaves on these subcategories with respect to the atomic topology are precisely the classifying toposes of the *completions* of the theory $\mathbb{T}'$, i.e. of the (complete) quotients of $\mathbb{T}'$ obtained by adding an axiom of the form $(\top \vdash_{\square} \phi)$ for a $\mathbb{T}'$ -complete geometric sentence ϕ .

Theorem 10.4.5 is a vast generalization of the well-known result (Theorem 7.4.1(a) [49]) allowing one to build countably categorical theories through Fraïssé's method (the classical result can be recovered as the particular case of the theorem when $\mathbb{T}$ is the quotient of the empty theory over a finite signature corresponding as in Corollary 9.1.3 to a uniformly finite collection of finitely presented models of it satisfying the hereditary property). In fact, as argued in Chapter 6, the context of theories of presheaf type is a very general and natural one both from a mathematical and a model-theoretic perspective; moreover, as we have seen in the previous chapters, there are many effective means for generating theories of presheaf type or testing whether a certain theory is of presheaf type.

Note that Theorem 10.4.5 represents a faithful application of the 'bridge' technique (cf. section 2.2) in the context of the canonical Morita equivalence induced by a theory of presheaf type, the invariants used in the proof being the notion of subtopos, the notion of geometric morphism to a classifying topos, the properties of a topos to be atomic and two-valued and the notion of atom of a topos:

$$\begin{array}{ccc}
 & [\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, \mathbf{Set}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) & \\
 & \uparrow & \uparrow \\
 \text{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) & & \\
 \downarrow & & \downarrow \\
 (\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, T) & & \mathbb{T} \\
 \downarrow & & \downarrow \\
 (\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) & & \mathbb{T}'
 \end{array}$$

The notion of homogeneous $\mathbb{T}$ -model is most relevant when all the $\mathbb{T}$ -model homomorphisms in $\mathbf{Set}$ are injective; the notion of injectivization of a geometric theory, which we introduced in section 7.2.2, thus plays an important role in this context.

Examples of theories to which Theorem 10.4.5 can be applied include the following:

- (a) The theory $\mathbb{I}$ of decidable objects (that is, the injectivization of the empty theory over a one-sorted signature, in the sense of Definition 7.2.10) is of presheaf type, and its category of finitely presentable models is the category $\mathbf{I}$ of finite sets and injections between them. The theory of homogeneous $\mathbb{I}$ -models is classified by the topos $\mathbf{Sh}(\mathbf{I}^{\text{op}}, J_{\text{at}})$, also known as the *Schanuel topos*. Notice that the homogeneous $\mathbb{I}$ -models in $\mathbf{Set}$ are precisely the infinite sets.
- (b) The theory of decidable Boolean algebras (that is, the injectivization of the algebraic theory of Boolean algebras) is of presheaf type (cf. section 9.1). Its finitely presentable models are the finite Boolean algebras, while its homogeneous set-based models are the atomless Boolean algebras. The well-known result that any two countable atomless Boolean algebras are isomorphic thus follows from Theorem 10.4.5 assuming the axiom of countable choice.
- (c) The theory of (decidably) linearly ordered objects (that is, the injectivization of the coherent theory of linear orders) is of presheaf type (cf. section 9.1), and its category of finitely presentable models coincides with the category of finite linear orders and order-preserving injections between them. Its homogeneous models are precisely the dense linearly ordered objects without endpoints (cf. section 8.1.1.1). Theorem 10.4.5 thus ensures that the theory of dense linearly ordered objects without endpoints is atomic and complete.
- (d) As shown in [29] (cf. Propositions 2.3 and 2.4 therein), the (infinitary) geometric theory of fields of finite characteristic which are algebraic over their prime field is of presheaf type and the theory of its homogeneous models can be identified with the theory of fields of finite characteristic which are algebraic over their prime field and algebraically closed; the completions of this theory are obtained precisely by adding, in each case, the axiom fixing the characteristic of the field (i.e. the sequent $(\top \vdash_{\square} p.1 = 0)$ for a prime number p). Theorem 10.4.5 thus implies, assuming the countable axiom of choice, the (well-known) fact that any two algebraic closures of a given finite field are isomorphic and, without assuming any form of the axiom of choice,

the fact that any two algebraic closures of a given finite field satisfy the same first-order sentences (cf. Remark 10.4.7) written in the language of fields; in fact, this property is true more generally for any base field (the theory of algebraic extensions of a base field is of presheaf type – cf. section 9.5 – and satisfies the hypotheses of Theorem 10.4.5).

Given a theory of presheaf type $\mathbb{T}$ whose category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ of finitely presentable models satisfies AP then, by Remark 8.1.9, the theory $\mathbb{T}'$ of homogeneous $\mathbb{T}$ -models can be identified with the *Booleanization* of the theory $\mathbb{T}$ defined in section 4.2.3, that is as the quotient of $\mathbb{T}$ obtained by adding the sequent

$$(\top \vdash_{\vec{y}} \psi)$$

for any $\mathbb{T}$ -stably consistent formula-in-context $\psi(\vec{y})$.

Remark 10.4.7 As every atomic topos is Boolean (since every atomic frame is isomorphic to a powerset), every atomic theory is Boolean. Every Boolean geometric theory satisfies the following key property: every first-order formula-in-context over its signature is provably equivalent in the theory, using classical logic, to a geometric formula in the same context (cf. Remark 4.2.19). This property has two immediate but remarkable consequences:

- Every Boolean geometric theory is complete as a geometric theory (in the sense of Definition 4.2.21) if and only if it is complete in the sense of classical model theory (that is, every first-order sentence over its signature is provably false or provably true using classical logic, but not both). In general, such an equivalence between the two notions of completeness only holds at the cost of replacing the given first-order theory with a geometric theory over a larger signature, e.g. its Morleyization (cf. Remark 2.1.3).
- Every Boolean geometric theory (in particular, every theory of homogeneous models as in Theorem 10.4.5) is model-complete (in the sense of classical model theory), that is every embedding of set-based models of it is an elementary embedding.

There is a kind of converse to Theorem 10.4.5, relating the amalgamation property on the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ with a semantic property of the theory $\mathbb{T}'$. In fact, the theory $\mathbb{T}'$ of homogeneous $\mathbb{T}$ -models can be defined even if $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}$ does not satisfy the right Ore condition, by replacing the atomic topology J_{at} with the Grothendieck topology on it generated by the family of all non-empty sieves.

Theorem 10.4.8 (Theorem 3.12 [23]). *Let $\mathbb{T}$ be a theory of presheaf type and $\mathbb{T}'$ the theory of homogeneous $\mathbb{T}$ -models. Suppose that $\mathbb{T}'$ has enough set-based models and that all the arrows in $\mathbb{T}\text{-mod}(\mathbf{Set})$ are monic. Then the following conditions are equivalent:*

- (i) *The category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ satisfies the amalgamation property.*

- (ii) For every finitely presentable $\mathbb{T}$ -model c there exist a set-based $\mathbb{T}'$ -model (that is, a homogeneous $\mathbb{T}$ -model in **Set**) M and a $\mathbb{T}$ -model homomorphism $c \rightarrow M$.

We can naturally expect the consideration of invariants other than the ones used in the proof of Theorem 10.4.5 in connection with the same Morita equivalence for $\mathbb{T}'$ to lead to other relationships between the geometry of the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ and the syntactic properties of the theory $\mathbb{T}'$. For example, the consideration of the invariant notion of arrow between two atoms leads to the following result:

Proposition 10.4.9 *Let $\mathbb{T}$ be a theory of presheaf type such that the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ satisfies the amalgamation property, and let $\mathbb{T}'$ be the theory of homogeneous $\mathbb{T}$ -models. Then, if all the arrows in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ are strict monomorphisms (in the sense of condition (i) of Theorem 10.5.5 below), every $\mathbb{T}'$ -provably functional geometric formula from a formula $\{\vec{x}. \phi\}$ which presents a $\mathbb{T}$ -model to another formula $\{\vec{y}. \psi\}$ which presents a $\mathbb{T}$ -model is $\mathbb{T}$ -provably functional from $\{\vec{x}. \phi\}$ to $\{\vec{y}. \psi\}$ (equivalently, corresponds to a $\mathbb{T}$ -model homomorphism $M_{\{\vec{y}. \psi\}} \rightarrow M_{\{\vec{x}. \phi\}}$).*

Proof Using the above notation, given two formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ presenting respectively $\mathbb{T}$ -models $M_{\{\vec{x}. \phi\}}$ and $M_{\{\vec{y}. \psi\}}$, the objects $l_{J_{\text{at}}}(M_{\{\vec{x}. \phi\}})$ and $y_{\mathbb{T}'}(\{\vec{x}. \phi\})$ (resp. $l_{J_{\text{at}}}(M_{\{\vec{y}. \psi\}})$ and $y_{\mathbb{T}'}(\{\vec{y}. \psi\})$) correspond to each other under the Morita equivalence

$$\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}).$$

Therefore the arrows $l_{J_{\text{at}}}(M_{\{\vec{x}. \phi\}}) \rightarrow l_{J_{\text{at}}}(M_{\{\vec{y}. \psi\}})$ in $\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}})$ correspond bijectively to the arrows $y_{\mathbb{T}'}(\{\vec{x}. \phi\}) \rightarrow y_{\mathbb{T}'}(\{\vec{y}. \psi\})$ in $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}).$ But these arrows in turn correspond, $y'_{\mathbb{T}}$ being full and faithful, to the arrows $\{\vec{x}. \phi\} \rightarrow \{\vec{y}. \psi\}$ in $\mathcal{C}_{\mathbb{T}'}$, that is to the $\mathbb{T}'$ -provably functional formulae $\{\vec{x}. \phi\} \rightarrow \{\vec{y}. \psi\}$.

Now, one can easily prove that the topology J_{at} is subcanonical if and only if every arrow in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ is a strict monomorphism (in the sense of satisfying condition (i) of Theorem 10.5.5). So, if this condition holds then the arrows $l_{J_{\text{at}}}(M_{\{\vec{x}. \phi\}}) \rightarrow l_{J_{\text{at}}}(M_{\{\vec{y}. \psi\}})$ in $\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}})$ correspond precisely to the $\mathbb{T}$ -model homomorphism $M_{\{\vec{y}. \psi\}} \rightarrow M_{\{\vec{x}. \phi\}}$ (equivalently, by Theorem 6.1.13, to the $\mathbb{T}$ -provably functional formulae $\{\vec{x}. \phi\} \rightarrow \{\vec{y}. \psi\}$). In other words, we have the following ‘bridge’:

$$\begin{array}{ccc} & \text{Arrows } [[\vec{x}. \phi]]_U \dashv \dashv [[\vec{y}. \psi]]_U & \\ & \mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) & \\ \swarrow & & \searrow \\ \begin{array}{l} (\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) \\ \text{Arrows } M_{\{\vec{y}. \psi\}} \rightarrow M_{\{\vec{x}. \phi\}} \\ \text{in f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}) \end{array} & & \begin{array}{l} (\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) \\ \mathbb{T}'\text{-provably functional formulae} \\ \{\vec{x}. \phi\} \rightarrow \{\vec{y}. \psi\} \end{array} \end{array}$$

(where U denotes ‘the’ universal model of $\mathbb{T}'$ in its classifying topos). $\square$

We shall reconsider the same invariant in section 10.5 in the context of Galois-type representations for the classifying topos of $\mathbb{T}'$ (cf. Theorem 10.5.3 and Remark 10.5.4(b)).

Another natural application of the ‘bridge technique’ in the context of the above Morita equivalence for $\mathbb{T}'$ is obtained by considering the invariant property of a topos to have enough points. It is interesting to consider this invariant in combination with the (invariant) property of a topos to be non-trivial (recall that a topos is said to be non-trivial if its initial and terminal objects are non-isomorphic) in the context of the investigation of the existence of points of a given topos; indeed, if a non-trivial topos has enough points then it has at least a point. Now, one can profitably exploit the geometry of sites of definition of a topos to investigate these kinds of issues. As an example, by Theorem 3.3 [6] the theory $\mathbb{T}^{\mathcal{C}}$ of flat functors on a small category $\mathcal{C}$ (cf. the proof of Theorem 2.1.11) can be axiomatized coherently over its signature if and only if $\mathcal{C}$ has all *finite fc-limits* (that is, for any finite diagram with values in $\mathcal{C}$ there is a finite family of cones over it such that every cone over the diagram factors through one in the family). Since $\mathbb{T}^{\mathcal{C}}$ is classified by the presheaf topos $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$, if the category $\mathcal{C}$ satisfies the right Ore condition then the theory obtained from $\mathbb{T}^{\mathcal{C}}$ by adding the axiom

$$(\top \vdash_{y^{\ulcorner d \urcorner}} (\exists x^{\ulcorner c \urcorner}) (\ulcorner f \urcorner(x) = y))$$

for any arrow $f : c \rightarrow d$ in $\mathcal{C}$, is classified by the topos $\mathbf{Sh}(\mathcal{C}, J_{\text{at}})$ and hence it is coherent if the theory $\mathbb{T}^{\mathcal{C}}$ is. So, if the category $\mathcal{C}$ has all finite fc-limits and satisfies the right Ore condition, the topos $\mathbf{Sh}(\mathcal{C}, J_{\text{at}})$ is coherent and hence, assuming the axiom of choice, it has enough points. Now, it is easy to see that the topos $\mathbf{Sh}(\mathcal{C}, J_{\text{at}})$ is non-trivial if and only if the category $\mathcal{C}$ is non-empty. Applying these remarks in connection with the Morita equivalence

$$\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'})$$

then yields the following existence theorem for homogeneous models:

Theorem 10.4.10 (Theorem 3.4 [23]). *Let $\mathbb{T}$ be a theory of presheaf type whose category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ of finitely presentable models satisfies the amalgamation property. If the topos $\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}})$ has enough points (e.g. if the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ has all finite fc-colimits) and there exists at least one $\mathbb{T}$ -model in $\mathbf{Set}$, then, assuming the axiom of choice, there exists at least one homogeneous $\mathbb{T}$ -model in $\mathbf{Set}$. $\square$*

Notice that this theorem represents a consistency result for logical theories arising from purely combinatorial/geometrical considerations.

10.5 Maximal theories and Galois representations

In this section, which is based on [27], we establish, under some natural hypotheses, Galois-type representations for the atomic two-valued toposes considered

in section 10.4 as toposes of continuous actions of a topological group of automorphisms of a suitable structure. This leads to a framework which generalizes Grothendieck's theory of Galois categories (cf. [47]) and allows one to build Galois-type theories in a great variety of new mathematical contexts.

Theorem 10.5.1 (cf. Corollary 3.7 [27]). *Let $\mathbb{T}$ be a theory of presheaf type whose category $\mathbf{f.p.T-mod}(\mathbf{Set})$ of finitely presentable models satisfies AP and JEP, and let M be a $\mathbf{f.p.T-mod}(\mathbf{Set})$ -universal and $\mathbf{f.p.T-mod}(\mathbf{Set})$ -ultrahomogeneous model of $\mathbb{T}$ (in the sense of Definition 10.4.3). Then we have an equivalence of toposes*

$$\mathbf{Sh}(\mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}}, J_{\mathrm{at}}) \simeq \mathbf{Cont}(\mathrm{Aut}(M)),$$

where $\mathrm{Aut}(M)$ is endowed with the topology of pointwise convergence (in which a basis of open neighbourhoods of the identity is given by the sets of the form $\{f : M \cong M \mid f(\vec{a}) = \vec{a}\}$ for a tuple $\vec{a}$ of elements of M), which is induced by the functor

$$F : \mathbf{f.p.T-mod}(\mathbf{Set})^{\mathrm{op}} \rightarrow \mathbf{Cont}(\mathrm{Aut}(M))$$

sending any model c of $\mathbf{f.p.T-mod}(\mathbf{Set})$ to the set $\mathrm{Hom}_{\mathbf{T-mod}(\mathbf{Set})}(c, M)$ (endowed with the obvious action by $\mathrm{Aut}(M)$) and any arrow $f : c \rightarrow d$ in $\mathbf{f.p.T-mod}(\mathbf{Set})$ to the $\mathrm{Aut}(M)$ -equivariant map

$$- \circ f : \mathrm{Hom}_{\mathbf{T-mod}(\mathbf{Set})}(d, M) \rightarrow \mathrm{Hom}_{\mathbf{T-mod}(\mathbf{Set})}(c, M).$$

Remark 10.5.2 Theorem 10.5.1 also admits a purely categorical formulation (not involving the concept of theory of presheaf type), obtained by replacing the category $\mathbf{f.p.T-mod}(\mathbf{Set})$ with an arbitrary small category $\mathcal{C}$, the category $\mathbf{T-mod}(\mathbf{Set})$ with the ind-completion $\mathrm{Ind}\text{-}\mathcal{C}$ of $\mathcal{C}$, the model M with an object u of $\mathrm{Ind}\text{-}\mathcal{C}$ and the group $\mathrm{Aut}(M)$ with the group of automorphisms of u in the category $\mathrm{Ind}\text{-}\mathcal{C}$, topologized in such a way that a basis of open neighbourhoods of the identity is given by the sets of the form $\{f : u \cong u \mid f \circ \chi = \chi\}$ for an arrow $\chi : c \rightarrow u$ in $\mathrm{Ind}\text{-}\mathcal{C}$ from an object c of $\mathcal{C}$ to u (see Theorem 3.5 [27]).

Under the hypotheses of Theorem 10.5.1, let $\mathbb{T}'$ be the theory of homogeneous $\mathbb{T}$ -models (as defined in section 10.4). Then the model M , endowed with the (continuous) canonical action of $\mathrm{Aut}(M)$, is a universal model of $\mathbb{T}'$ in the topos $\mathbf{Cont}(\mathrm{Aut}(M))$ (cf. the proof of Theorem 3.1 [27]). So for any tuple $A_1, \dots, A_n$ of sorts of the signature of $\mathbb{T}$, we have a ‘bridge’

$$\begin{array}{ccc} & \text{Subobject of } UA_1 \times \dots \times UA_n & \\ & \mathbf{Cont}(\mathrm{Aut}(M)) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) & \\ \swarrow & & \searrow \\ \begin{array}{l} \mathrm{Aut}(M) \\ \text{Aut}(M)\text{-closed subset} \\ S \subseteq MA_1 \times \dots \times MA_n \end{array} & & \begin{array}{l} (\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}) \\ \text{Geometric formula} \\ \phi(x_1^{A_1}, \dots, x_n^{A_n}) \end{array} \end{array}$$

(where U is ‘the’ universal model of $\mathbb{T}'$ in its classifying topos), which yields the following result:

Theorem 10.5.3 *Let $\mathbb{T}'$ be the theory of homogeneous $\mathbb{T}$ -models for a theory $\mathbb{T}$ satisfying the hypotheses of Theorem 10.5.1.*

- (i) *For any subset $S \subseteq MA_1 \times \cdots \times MA_n$ which is closed under the action of $\text{Aut}(M)$, there exists a (unique up to $\mathbb{T}'$ -provable equivalence) geometric formula $\phi(\vec{x})$ over the signature of $\mathbb{T}$ (where $\vec{x} = (x_1^{A_1}, \dots, x_n^{A_n})$) such that $S = [[\vec{x} \cdot \phi]]_M$.*
- (ii) *For any $\text{Aut}(M)$ -equivariant map $f : S \rightarrow T$ between $\text{Aut}(M)$ -closed subsets S and T as in (i) there exists a (unique up to $\mathbb{T}'$ -provable equivalence) $\mathbb{T}'$ -provably functional geometric formula $\theta(\vec{x}, \vec{y})$ from $\phi(\vec{x})$ to $\psi(\vec{y})$, where $S = [[\vec{x} \cdot \phi]]_M$ and $T = [[\vec{y} \cdot \psi]]_M$, whose interpretation $[[\vec{x}, \vec{y} \cdot \theta]]_M$ coincides with the graph of f .*

Remarks 10.5.4 (a) It easily follows from Theorem 10.5.3 that for any finite tuple $A_1, \dots, A_n$ of sorts of the signature of $\mathbb{T}$, the orbits of the action of $\text{Aut}(M)$ on $MA_1 \times \cdots \times MA_n$ coincide with the interpretations $[[\vec{x} \cdot \phi]]_M$ in M of $\mathbb{T}$ -complete formulae $\phi(\vec{x})$, where $\vec{x} = (x_1^{A_1}, \dots, x_n^{A_n})$, and hence correspond precisely to the $\mathbb{T}$ -provable equivalence classes of $\mathbb{T}$ -complete formulae in the context $\vec{x}$ (in the sense of Definition 10.3.1(a)).

- (b) If in part (ii) of Theorem 10.5.3 the formulae $\phi(\vec{x})$ and $\psi(\vec{y})$ present respectively $\mathbb{T}$ -models $M_{\{\vec{x} \cdot \phi\}}$ and $M_{\{\vec{y} \cdot \psi\}}$ and all the arrows in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ are strict monomorphisms (cf. Theorem 10.5.5 below), then the formula θ can be taken to be $\mathbb{T}$ -provably functional and hence to induce (by Theorem 6.1.13) a $\mathbb{T}$ -model homomorphism $z : M_{\{\vec{y} \cdot \psi\}} \rightarrow M_{\{\vec{x} \cdot \phi\}}$ such that the map $[[\vec{x} \cdot \phi]]_M \rightarrow [[\vec{y} \cdot \psi]]_M$ whose graph is $[[\vec{x}, \vec{y} \cdot \theta]]_M$ corresponds, under the identifications

$$\text{Hom}_{\mathbb{T}\text{-mod}(\mathbf{Set})}(M_{\{\vec{x} \cdot \phi\}}, M) \cong [[\vec{x} \cdot \phi]]_M$$

and

$$\text{Hom}_{\mathbb{T}\text{-mod}(\mathbf{Set})}(M_{\{\vec{y} \cdot \psi\}}, M) \cong [[\vec{y} \cdot \psi]]_M,$$

to the map $\text{Hom}_{\mathbb{T}\text{-mod}(\mathbf{Set})}(z, M)$.

Concerning the existence of models M satisfying the hypotheses of Theorem 10.5.1, we should remark that ultrahomogeneous structures naturally arise in many different mathematical contexts. Their existence can be proved either directly by means of an explicit construction or through abstract logical arguments. A general method for building countable ultrahomogeneous structures is provided by Fraïssé's construction in model theory (cf. Chapter 7 of [48]), while the categorical generalization established in section 2 of [23] allows one to construct ultrahomogeneous structures of arbitrary cardinality. As examples of models M satisfying the hypotheses of Theorem 10.5.1 in relation to the theories of presheaf type considered in section 10.4, we mention:

- The set $\mathbb{N}$ of natural numbers, with respect to the theory of decidable objects;

- The ordered set $(\mathbb{Q}, <)$ of rational numbers, with respect to the theory $\mathbb{L}$ of decidable linearly ordered objects considered in section 8.1.1.1;
- The unique countable atomless Boolean algebra, with respect to the theory of decidable Boolean algebras.
- Any Galois extension F' of a given field F , with respect to the $\mathcal{L}_F^{F'}$ -completion (in the sense of Theorem 8.2.10) of the theory $\mathbb{T}_F$ of algebraic extensions of F (defined in section 9.5), where $\mathcal{L}_F^{F'}$ is the category of finite extensions of F which are embeddable in F' and field homomorphisms between them.

There is a natural link between ultrahomogeneity and the property of a model of an atomic complete theory to be *special*, which can be exploited to construct models satisfying the hypotheses of Theorem 10.5.1. Recall that a model M of an atomic complete theory $\mathbb{T}$ is said to be *special* if every $\mathbb{T}$ -complete formula $\phi(\vec{x})$ is realized in M and, for any tuples $\vec{a}$ and $\vec{b}$ of elements of M which satisfy the same $\mathbb{T}$ -complete formula, there is an automorphism of M which sends $\vec{a}$ to $\vec{b}$. Any model M satisfying the hypotheses of Theorem 10.5.1 is clearly special as a model of the theory of homogeneous $\mathbb{T}$ -models. Conversely, given a special model M of an atomic complete theory $\mathbb{S}$, we have, by the Comparison Lemma (cf. Theorem 1.1.8), an equivalence $\mathbf{Sh}(\mathcal{C}_{\mathbb{S}}, J_{\mathbb{S}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{S}}^{\text{at}}, J_{\text{at}})$, where $\mathcal{C}_{\mathbb{S}}^{\text{at}}$ is the full subcategory of $\mathcal{C}_{\mathbb{S}}$ on the $\mathbb{S}$ -complete formulae, which shows that $\mathbb{S}$ is Morita-equivalent to the theory of homogeneous $\mathbb{T}$ -models for a theory of presheaf type $\mathbb{T}$ with respect to which the model M satisfies the hypotheses of Theorem 10.5.1.

Let us now discuss under which conditions the Morita equivalence of Theorem 10.5.1 restricts to an equivalence of sites.

Given an arrow $\chi : c \rightarrow M$ in $\mathbb{T}\text{-mod}(\mathbf{Set})$, where c is in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$, we denote by

$$\mathcal{I}_{\chi} := \{f : M \cong M \mid f \circ \chi = \chi\}$$

the open subgroup of $\text{Aut}(M)$ consisting of the automorphisms which fix χ .

The following theorem gives necessary and sufficient conditions for the functor F of Theorem 10.5.1 to be full and faithful.

Theorem 10.5.5 (cf. Proposition 4.1 [27]). *Under the hypotheses of Theorem 10.5.1, the following conditions are equivalent:*

- (i) *Every arrow $f : d \rightarrow c$ in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ is a strict monomorphism, i.e. for any arrow $g : e \rightarrow c$ in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ such that $h \circ g = k \circ g$ whenever $h \circ f = k \circ f$, g factors uniquely through f :*

$$\begin{array}{ccc} d & \xrightarrow{f} & c \\ \uparrow \lambda & \nearrow g & \xrightarrow[k]{h} u \\ | & & \\ | & & \\ e. & & \end{array}$$

- (ii) *The functor*

$$F : \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}} \rightarrow \mathbf{Cont}(\text{Aut}(M))$$

of Theorem 10.5.1 is full and faithful.

- (iii) For any models $c, d \in \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ and any arrows $\chi : c \rightarrow M$ and $\xi : d \rightarrow M$ in $\mathbb{T}\text{-mod}(\mathbf{Set})$, $\mathcal{I}_\xi \subseteq \mathcal{I}_\chi$ (that is, for any automorphism f of M , $f \circ \xi = \chi$ implies $f \circ \chi = \xi$) if and only if there exists exactly one arrow $f : c \rightarrow d$ in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ such that $\chi = \xi \circ f$:

$$\begin{array}{ccc} c & \xrightarrow{\chi} & M \\ \downarrow f & \nearrow \xi & \\ d & & \end{array}$$

- (iv) The Grothendieck topology J_{at} is subcanonical.

Let $\text{Sgr}(\text{Aut}(M))$ be the preorder category consisting of the open subgroups of $\text{Aut}(M)$ and the inclusion ordering between them, and let $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}/M$ be the category whose objects are the arrows $c \rightarrow M$ in $\mathbb{T}\text{-mod}(\mathbf{Set})$, where c is an object of $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$, and whose arrows $(\chi : c \rightarrow M) \rightarrow (\xi : d \rightarrow M)$ are the arrows $f : d \rightarrow c$ in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ such that $\chi \circ f = \xi$. Then the functor F of Theorem 10.5.1 yields a functor

$$\tilde{F} : \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}/M \rightarrow \text{Sgr}(\text{Aut}(M))$$

sending any object $\chi : c \rightarrow M$ of $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}/M$ to the subgroup $\mathcal{I}_\chi$ of $\text{Aut}(M)$.

Note that if the conditions of Theorem 10.5.5 are satisfied then the slice category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}/M$ is a preorder. Indeed, if all the arrows of the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ are monic then all the arrows of the category $\mathbb{T}\text{-mod}(\mathbf{Set})$ are monic too (cf. the proof of Theorem 7.2.32).

It is natural to wonder under which conditions the functor $\tilde{F}$ yields a bijection between the objects of the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}/M$ and the open subgroups of the topological group $\text{Aut}(M)$, as in the case of classical Galois theory. The following theorem provides an answer to this question.

Theorem 10.5.6 (cf. Theorem 4.17 [27]). *Assuming that the equivalent conditions of Theorem 10.5.5 are satisfied, the following conditions are equivalent:*

- (i) The functor

$$F : \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}} \rightarrow \mathbf{Cont}(\text{Aut}(M))$$

is a categorical equivalence onto the full subcategory of $\mathbf{Cont}(\text{Aut}(M))$ on the non-empty transitive actions.

- (ii) The map

$$\tilde{F} : \text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}/M \rightarrow \text{Sgr}(\text{Aut}(M))$$

is a bijection onto the set of open subgroups of $\text{Aut}(M)$.

- (iii) The category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ has equalizers and arbitrary intersections of subobjects of a given object and for any pair of arrows $h, k : c \rightarrow e$ in $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ with equalizer $m : d \rightarrow c$ we have that for any pair of arrows $l, n : c \rightarrow e'$, $l \circ m = n \circ m$ if and only if there exists an arrow $s : e' \rightarrow e''$ such that $(s \circ l, s \circ n)$ belongs to the equivalence relation on $\text{Hom}_{\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})}(c, e'')$ generated by the relation consisting of the pairs of the form $(t \circ h, t \circ k)$ for an arrow $t : e \rightarrow e''$.
- (iv) Every atom of the topos $\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}})$ is, up to isomorphism, of the form $l_{J_{\text{at}}}(c)$ for some object c of $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$, where $l_{J_{\text{at}}}$ is the functor

$$\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set}) \rightarrow \mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{\text{op}}, J_{\text{at}})$$

given by the composite of the Yoneda embedding with the associated sheaf functor.

Remark 10.5.7 If the category $\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})$ satisfies the hypotheses of Theorem 10.5.6, it is always possible to ‘complete’ it to a category satisfying the equivalent conditions of the theorem and having an equivalent associated topos, by means of an elementary process which is a sort of completion by the addition of imaginaries (in the sense of model theory). This process is described in detail in [27] (cf. the proof of Theorem 4.15 therein) and applies to a number of classical categories such as the category of Boolean algebras and embeddings, the category of finite groups and embeddings, the category of finite graphs and embeddings, etc.

Notice that both Theorem 10.5.5 and Theorem 10.5.6 naturally arise from topos-theoretic ‘bridges’, the former by considering the invariant notion of arrow between two given objects, and the latter by considering the notion of atom of a topos.

10.6 A characterization theorem for geometric logic

As an application of the duality theorem of Chapter 3, we can derive a criterion for deciding whether a class of structures is the class of models of a geometric theory inside Grothendieck toposes; this criterion has led in particular to a characterization of the infinitary first-order theories which are geometric in terms of their models in Grothendieck toposes, solving a problem posed by I. Moerdijk in 1989 (cf. [15]).

The proof of our criterion relies on the following technical lemma:

Lemma 10.6.1 (Lemma 1 [15]). *Let $\{f_i : \mathcal{E}_i \rightarrow \mathcal{E} \mid i \in I\}$ be a family of geometric morphisms of (elementary) toposes with common codomain $\mathcal{E}$. Then $\{f_i : \mathcal{E}_i \rightarrow \mathcal{E} \mid i \in I\}$ is jointly surjective if and only if $\mathcal{E} = \bigcup_{i \in I} \mathcal{E}'_i$, where for each $i \in I$, $\mathcal{E}_i \twoheadrightarrow \mathcal{E}'_i \hookrightarrow \mathcal{E}$ is the surjection-inclusion factorization of f_i .*

Theorem 10.6.2 (cf. Theorem 2 [15]). *Let Σ be a first-order signature and $\mathcal{S}$ a collection of Σ -structures in Grothendieck toposes closed under isomorphisms*

of structures. Then $\mathcal{S}$ is the collection of all models in Grothendieck toposes of a geometric theory over Σ if and only if it satisfies the following two conditions:

- (i) For any geometric morphism $f : \mathcal{F} \rightarrow \mathcal{E}$, if $M \in \Sigma\text{-str}(\mathcal{E})$ is in $\mathcal{S}$ then $f^*(M)$ is in $\mathcal{S}$.
- (ii) For any (set-indexed) jointly surjective family $\{f_i : \mathcal{E}_i \rightarrow \mathcal{E} \mid i \in I\}$ of geometric morphisms and any Σ -structure M in $\mathcal{E}$, if $f_i^*(M)$ is in $\mathcal{S}$ for every $i \in I$ then M is in $\mathcal{S}$.

Proof The ‘only if’ part of the theorem is trivial and well-known. Let us prove the ‘if’ part. Let $\mathbb{O}_\Sigma$ be the empty (geometric) theory over Σ . The $\mathbb{O}_\Sigma$ -models in any Grothendieck topos $\mathcal{E}$ are precisely the Σ -structures in $\mathcal{E}$ and hence, for any Grothendieck topos $\mathcal{E}$, the geometric morphisms $\mathcal{E} \rightarrow \mathbf{Set}[\mathbb{O}_\Sigma]$ correspond to the Σ -structures in $\mathcal{E}$. The geometric morphism corresponding to a Σ -structure M will be denoted by $f_M : \mathcal{E}_M \rightarrow \mathbf{Set}[\mathbb{O}_\Sigma]$; recall that if U is a universal model of $\mathbb{O}_\Sigma$ in $\mathbf{Set}[\mathbb{O}_\Sigma]$ then $M \cong f_M^*(U)$.

Let $\mathcal{E} \hookrightarrow \mathbf{Set}[\mathbb{O}_\Sigma]$ be the subtopos of $\mathbf{Set}[\mathbb{O}_\Sigma]$ given by the union of all the subtoposes of $\mathbf{Set}[\mathbb{O}_\Sigma]$ arising as the inclusion parts of the surjection-inclusion factorizations of the geometric morphisms $f_M : \mathcal{E}_M \rightarrow \mathbf{Set}[\mathbb{O}_\Sigma]$ for M in $\mathcal{S}$, and let $a : \mathbf{Set}[\mathbb{O}_\Sigma] \rightarrow \mathcal{E}$ be the corresponding associated sheaf functor. By Theorem 3.2.5 the subtopos $\mathcal{E} \hookrightarrow \mathbf{Set}[\mathbb{O}_\Sigma]$ of $\mathbf{Set}[\mathbb{O}_\Sigma]$ corresponds to a (unique up to syntactic equivalence) geometric quotient $\mathbb{T}$ of $\mathbb{O}_\Sigma$ such that if $U_{\mathbb{O}_\Sigma}$ is a universal model of $\mathbb{O}_\Sigma$ in $\mathbf{Set}[\mathbb{O}_\Sigma]$ then $U_{\mathbb{T}} := a(U_{\mathbb{O}_\Sigma})$ is a universal model of $\mathbb{T}$ in $\mathcal{E}$. Let us show that $\mathbb{T}$ axiomatizes our class of structures $\mathcal{S}$.

For any structure $M \in \Sigma\text{-str}(\mathcal{E}_M)$ in $\mathcal{S}$, the subtopos $\mathcal{E}'_M \hookrightarrow \mathbf{Set}[\mathbb{O}_\Sigma]$ arising in the surjection-inclusion factorization of $f_M : \mathcal{E}_M \rightarrow \mathbf{Set}[\mathbb{O}_\Sigma]$ factors as the canonical inclusion $\mathcal{E}'_M \hookrightarrow \mathcal{E}$ followed by the inclusion $\mathcal{E} \hookrightarrow \mathbf{Set}[\mathbb{O}_\Sigma]$. Now, composing the former inclusion with the surjection part of the surjection-inclusion factorization of f_M yields a geometric morphism $h_M : \mathcal{E}_M \rightarrow \mathcal{E}$ which, when composed with $\mathcal{E} \hookrightarrow \mathbf{Set}[\mathbb{O}_\Sigma]$, is isomorphic to f_M . But $M \cong f_M^*(U_{\mathbb{O}_\Sigma})$, from which it follows that $h_M^*(U_{\mathbb{T}}) \cong M$ and hence that M is a model of $\mathbb{T}$. This shows that every structure in $\mathcal{S}$ is a model of $\mathbb{T}$. Let us prove that, conversely, any $\mathbb{T}$ -model in a Grothendieck topos lies in $\mathcal{S}$. By Lemma 10.6.1 the family of geometric morphisms h_M for M in $\mathcal{S}$ is jointly surjective; so condition (ii) implies that $U_{\mathbb{T}}$ lies in $\mathcal{S}$. Now, by the universal property of the classifying topos $\mathcal{E}$ of $\mathbb{T}$, every model N of $\mathbb{T}$ in a Grothendieck topos $\mathcal{F}$ is of the form $g^*(U_{\mathbb{T}})$ for some geometric morphism $g : \mathcal{F} \rightarrow \mathcal{E}$, so condition (i) implies our thesis.

Note in passing that $\mathbb{T}$ can be described as the collection of geometric sequents over Σ which are valid in every structure M in $\mathcal{S}$ (cf. Theorem 4.2.13). $\square$

10.7 The maximal spectrum of a commutative ring

The purpose of this section is to show that, by using the ‘bridge’ technique (cf. section 2.2), we can design natural analogues of the Zariski spectrum of a commutative ring with unit which may be useful in different contexts, for

instance in connection with the problem of building an associated theory for the maximal spectrum of a ring.

As shown in [16], the different constructions of the Zariski spectrum of a commutative ring with unit A can be interpreted topos-theoretically as different representations of the topos of sheaves on it. For instance, we have a Morita equivalence

$$\mathbf{Sh}(\mathrm{Spec}(A)) \simeq \mathbf{Sh}(\mathrm{Rad}(A), J_A),$$

where $\mathrm{Rad}(A)$ is the frame of radical ideals of A (with the subset-inclusion ordering) and J_A is the canonical topology on it, and a Morita equivalence

$$\mathbf{Sh}(\mathrm{Spec}(A)) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{P}_A}, J_{\mathbb{P}_A}),$$

where $(\mathcal{C}_{\mathbb{P}_A}, J_{\mathbb{P}_A})$ is the geometric syntactic site of the theory $\mathbb{P}_A$ of prime filters on A defined in section 2.1.6.1. In fact, the Zariski spectrum of A is the space of points of the classifying topos of $\mathbb{P}_A$ (in the sense of Definition 1.1.23).

The first Morita equivalence can be established by using the well-known isomorphism between the frame $\mathcal{O}(\mathrm{Spec}(A))$ of open sets of $\mathrm{Spec}(A)$ and the frame $\mathrm{Rad}(A)$, which sends every radical ideal I of A to the complement in $\mathrm{Spec}(A)$ of the set $\{P \in \mathrm{Spec}(A) \mid P \supseteq I\}$.

The second Morita equivalence follows from the prime ideal theorem for distributive lattices (a theorem which is equivalent to a weak form of the axiom of choice), which ensures that, the theory $\mathbb{P}_A$ being coherent, its classifying topos has enough points and hence can be represented topologically as the topos of sheaves on its space of points.

The syntactic method for constructing classifying toposes of coherent theories reviewed in section 2.1.2 yields a further representation

$$\mathbf{Sh}(\mathrm{Spec}(A)) \simeq \mathbf{Sh}(L(A), J_A^{\mathrm{coh}})$$

of the topos $\mathbf{Sh}(\mathrm{Spec}(A))$ as the topos of coherent sheaves on a distributive lattice, namely the coherent syntactic category $L(A)$ of the propositional theory $\mathbb{P}_A$.

By the Comparison Lemma (Theorem 1.1.8), the site $(L(A), J_A^{\mathrm{coh}})$ can be cut down to a smaller site $(S(A), C_A)$ generating the same topos (up to equivalence), given by the full subcategory $S(A)$ of $L(A)$ on the elements of the form $[a]$ for $a \in A$, with the induced topology C_A . This site can be intrinsically characterized as follows: $S(A)$ is the meet-semilattice given by the quotient of the underlying monoid of A by the smallest congruence which identifies a and a^2 for all elements a of A , with the order given by $[a] \leq [b]$ if and only if $a \cdot b = a$ (where $[a]$ denotes the equivalence class of a in $S(A)$), while C_A is the Grothendieck topology on $S(A)$ generated by the covering families of the form $\emptyset \in C_A([0_A])$ and of the form $[a_i] \rightarrow [a]$ (for $i = 1, \dots, n$), where $[a_i] \leq [a]$ and $[a_1] + \dots + [a_n] = [a] = x$ (cf. sections 8.7 of [16] and V3.1 of [53]).

Each of these different representations of the small Zariski topos of A yields a different point of view on it; depending on the aspects of the spectrum that

one wishes to investigate, one point of view can be more natural or convenient than another.

Now that we have reviewed the topos-theoretic interpretation of the different constructions of the Zariski spectrum of A , let us proceed to discuss how to design a suitable analogue of it for the set of maximal ideals of A .

Since every maximal ideal is prime, it is natural to put a topology on the set $\text{Max}(A)$ of maximal ideals of A such that the inclusion $\text{Max}(A) \hookrightarrow \text{Spec}(A)$ induces a geometric inclusion $\mathbf{Sh}(\text{Max}(A)) \rightarrow \mathbf{Sh}(\text{Spec}(A))$. Now, for the inclusion map $\text{Max}(A) \hookrightarrow \text{Spec}(A)$ to be continuous it is necessary for the topology on $\text{Max}(A)$ to contain the subspace topology. On the other hand, any subspace inclusion $i : Y \hookrightarrow X$ induces a geometric inclusion $\mathbf{Sh}(i) : \mathbf{Sh}(Y) \hookrightarrow \mathbf{Sh}(X)$ (cf. Proposition C1.5.1(i) [55]). So, if we endow $\text{Max}(A)$ with the topology induced by that of $\text{Spec}(A)$, the topos $\mathbf{Sh}(\text{Max}(A))$ gets naturally identified with a subtopos of $\mathbf{Sh}(\text{Spec}(A))$, which allows us to ‘localize’ all the representations of $\mathbf{Sh}(\text{Spec}(A))$ to representations of $\mathbf{Sh}(\text{Max}(A))$. We shall now proceed to describe and analyse in detail such localizations, starting from the logical representation

$$\mathbf{Sh}(\text{Spec}(A)) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{P}_A}, J_{\mathbb{P}_A}).$$

By Theorem 3.2.5, there exists exactly one quotient $\mathbb{P}_A^{\max}$ of $\mathbb{P}_A$ classified by the subtopos $\mathbf{Sh}(\text{Max}(A)) \hookrightarrow \mathbf{Sh}(\text{Spec}(A))$; that is, we have a ‘bridge’

$$\begin{array}{ccccc}
 & \mathbf{Sh}(\text{Spec}(A)) & \simeq & \mathbf{Sh}(\mathcal{C}_{\mathbb{P}_A}, J_{\mathbb{P}_A}) & \\
 & \uparrow & & \uparrow & \\
 & \mathbf{Sh}(\text{Max}(A)) & \simeq & \mathbf{Sh}(\mathcal{C}_{\mathbb{P}_A^{\max}}, J_{\mathbb{P}_A^{\max}}) & \\
 \text{Spec}(A) & \text{---} & & & \text{---} & \mathbb{P}_A \\
 \uparrow & & & & & \downarrow \\
 \text{Max}(A) & & & & & \mathbb{P}_A^{\max}.
 \end{array}$$

By considering the topos-theoretic invariant notion of point of a topos in connection with the Morita equivalence $\mathbf{Sh}(\text{Max}(A)) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{P}_A^{\max}}, J_{\mathbb{P}_A^{\max}})$ and recalling the well-known site characterization ‘the points of the topos $\mathbf{Sh}(X)$ of sheaves on a topological space X correspond precisely to the points of the sobrification of X ’ (holding for any X), we obtain a ‘bridge’ yielding a bijective correspondence between the set-based models of the theory $\mathbb{P}_A^{\max}$ and the elements of the sobrification of the space $\text{Max}(A)$:

$$\begin{array}{ccc}
 & \text{Point of} & \\
 & \mathbf{Sh}(\text{Max}(A)) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{P}_A^{\max}}, J_{\mathbb{P}_A^{\max}}) & \\
 \text{Max}(A) & \text{---} & \mathbb{P}_A^{\max} \\
 \text{Point of the sobrification of Max}(A) & & \text{Set-based model of } \mathbb{P}_A^{\max}
 \end{array}$$

It follows in particular that the sobrification of the space $\text{Max}(A)$ can be realized as a space of prime ideals of A ; we shall characterize these ideals more explicitly

in Theorem 10.7.1. Another consequence of the above Morita equivalence is the fact that every open set of $\text{Max}(A)$ is of the form $\{M \in \text{Max}(A) \mid M \models \phi\}$ for a geometric sentence ϕ over the signature of the theory $\mathbb{P}_A$.

By considering the topos-theoretic invariant property ‘to be a dense subobject with respect to a subtopos’ in connection with the same Morita equivalence, we obtain a syntactic description of the theory $\mathbb{P}_A^{\max}$, as follows. Clearly, a geometric sequent $(\phi \vdash_{\vec{x}} \psi)$ over the signature of $\mathbb{P}_A$ is provable in $\mathbb{P}_A^{\max}$ if and only if the subobject $[[\vec{x} \cdot \phi \wedge \psi]]_U \rightarrowtail [[\vec{x} \cdot \phi]]_U$, where U is ‘the’ universal model of $\mathbb{P}_A$ in $\mathbf{Sh}(\mathcal{C}_{\mathbb{P}_A}, J_{\mathbb{P}_A})$, is dense with respect to the subtopos $\mathbf{Sh}(\mathcal{C}_{\mathbb{P}_A^{\max}}, J_{\mathbb{P}_A^{\max}})$. Now, since the points of the topos $\mathbf{Sh}(\text{Max}(A))$ corresponding to the maximal ideals of A are jointly surjective (that is, their inverse image functors jointly reflect isomorphisms), this subobject is dense if and only if the sequent $(\phi \vdash_{\vec{x}} \psi)$ is satisfied in every prime filter of A whose complement is a maximal ideal of A . We can thus conclude that the quotient $\mathbb{P}_A^{\max}$ is obtained from the theory $\mathbb{P}_A$ by adding all the sequents over the signature of $\mathbb{P}_A$ which are satisfied by all the complements of the maximal ideals of A .

Let us now consider the problem of getting an algebraic presentation of the maximal spectrum of A , as defined above; this will lead in particular to an alternative axiomatization of the theory $\mathbb{P}_A^{\max}$. By considering the invariant notion of subtopos in connection with the Morita equivalence

$$\mathbf{Sh}(\text{Spec}(A)) \simeq \mathbf{Sh}(L(A), J_A^{\text{coh}}),$$

we obtain that the subtopos $\mathbf{Sh}(\text{Max}(A)) \hookrightarrow \mathbf{Sh}(\text{Spec}(A))$ corresponds to a unique Grothendieck topology $J_A^{\max}$ on $L(A)$ such that the canonical geometric inclusion

$$\mathbf{Sh}(L(A), J_A^{\max}) \hookrightarrow \mathbf{Sh}(L(A), J_A^{\text{coh}})$$

induces an equivalence of subtoposes

$$\begin{array}{ccc} \mathbf{Sh}(\text{Spec}(A)) & \simeq & \mathbf{Sh}(L(A), J_A^{\text{coh}}) \\ \uparrow & & \uparrow \\ \mathbf{Sh}(\text{Max}(A)) & \simeq & \mathbf{Sh}(L(A), J_A^{\max}). \end{array}$$

Theorem 3.2.5 then implies, in view of the Morita equivalence

$$\mathbf{Sh}(\mathcal{C}_{\mathbb{P}_A}, J_{\mathbb{P}_A}) \simeq \mathbf{Sh}(S(A), C_A),$$

that the quotient $\mathbb{P}_A^{\max}$ is obtained by adding all the sequents over the signature of $\mathbb{P}_A$ of the form

$$\left(P_c \vdash \bigvee_{i \in I} P_{c_i} \right)$$

for $J_A^{\max}$ -covering sieves $\{[P_{c_i}] \rightarrow [P_c] \mid i \in I\}$ (notice that for each $a \in A$, $[a] \in S(A)$ and $[P_a] \in L(A)$ correspond to each other under the above equivalence $\mathbf{Sh}(L(A), J_A^{\text{coh}}) \simeq \mathbf{Sh}(S(A), C_A)$).

An explicit description of the topology $J_A^{\max}$ will thus provide us with an alternative axiomatization of the theory $\mathbb{P}_A^{\max}$. In order to obtain such a description, we shall use the characterization of the J -covering sieves S on a small category $\mathcal{C}$ (for a Grothendieck topology J on $\mathcal{C}$) as the subobjects $S \rightarrow \mathcal{C}(-, c)$ which are dense with respect to the closure operation on $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ associated with J . To calculate the closure operation on $\mathbf{Sh}(\text{Spec}(A))$ corresponding to the geometric inclusion $\mathbf{Sh}(\text{Max}(A)) \hookrightarrow \mathbf{Sh}(\text{Spec}(A))$, we observe that for any subspace Y of a topological space X , the action on subterminals of the inverse image of the subtopos inclusion $\mathbf{Sh}(Y) \hookrightarrow \mathbf{Sh}(X)$ can be identified with the map sending any open set U of X to the largest open set of X whose intersection with Y is contained in U . Next, we observe that, since the maximal ideals of A can be characterized in an invariant way as the points of the Zariski topos $\mathbf{Sh}(\text{Spec}(A))$ which are minimal with respect to the specialization ordering, and the points of the topos $\mathbf{Sh}(L(A), J_A^{\text{coh}})$ correspond to the prime ideals of the lattice $L(A)$, the Morita equivalence

$$\mathbf{Sh}(\text{Spec}(A)) \simeq \mathbf{Sh}(L(A), J_A^{\text{coh}})$$

restricts to an equivalence

$$\mathbf{Sh}(\text{Max}(A)) \simeq \mathbf{Sh}(\text{Max}(L(A))),$$

where $\text{Max}(L(A))$ is the subspace of the prime spectrum $\text{Spec}(L(A))$ of $L(A)$ on the maximal ideals of $L(A)$.

Now, for any distributive lattice D , the subspace $\text{Max}(D)$ of the prime spectrum $\text{Spec}(D)$ of D consisting of the maximal ideals of D corresponds to a unique Grothendieck topology $J_D^{\max}$ on D yielding an equivalence of subtoposes

$$\begin{array}{ccc} \mathbf{Sh}(\text{Spec}(D)) & \simeq & \mathbf{Sh}(D, J_D^{\text{coh}}) \\ \uparrow & & \uparrow \\ \mathbf{Sh}(\text{Max}(D)) & \simeq & \mathbf{Sh}(D, J_D^{\max}). \end{array}$$

For any ideal I of D , the open subset of $\text{Spec}(D)$ corresponding to it is given by $\{P \in \text{Spec}(D) \mid P \not\supseteq I\}$. The closure operation on $\mathbf{Sh}(D, J_D^{\text{coh}})$ corresponding to the geometric inclusion $\mathbf{Sh}(D, J_D^{\max}) \hookrightarrow \mathbf{Sh}(D, J_D^{\text{coh}})$ can thus be described, in terms of ideals of D , as the assignment sending any ideal I of D to the intersection of all the maximal ideals of D containing I . Under the hypothesis of the maximal ideal theorem for D (i.e. the assertion that every non-trivial ideal of D is contained in a maximal ideal), this can in turn be identified with the map sending any ideal I of D to the ideal

$$\{d \in D \mid (\forall b \in D)(b \vee d = 1) \Rightarrow (\exists c \in I)(b \vee c = 1)\}$$

(cf. section 1 of [54]). Indeed, for any element d of D , d does not belong to the intersection of all the maximal ideals of D containing I if and only if there exist a maximal ideal M of D containing I and an element $c \in M$ such that

$c \vee d = 1$; in other words, d belongs to the intersection of all the maximal ideals of D containing I if and only if for every element $b \in D$, $b \vee d = 1$ implies that for every maximal ideal M of D containing I , $b \notin M$. But, by the maximal ideal theorem for D , this latter condition means precisely that the ideal generated by b and I is trivial i.e. that there exists an element $c \in I$ such that $b \vee c = 1$.

This characterization yields the following explicit description of the topology $J_D^{\max}$: a sieve $\{c_i \rightarrow c \mid i \in I\}$ on $c \in D$ is $J_D^{\max}$ -covering if and only if for every $d \in D$, $d \vee c = 1$ implies that there exists a finite subset $J \subseteq I$ such that $\bigvee_{i \in J} c_i \vee d = 1$. Notice in particular that every $J_D^{\max}$ -covering sieve on the top element 1_D of D is generated by a finite family of arrows (take $d = 0$). This implies, in view of Proposition 10.8.1(iv), that (assuming the maximal ideal theorem for D) the space $\text{Max}(D)$ is compact:

$$\begin{array}{ccc} & \text{Sh}(D, J_D^{\max}) \simeq \text{Sh}(\text{Max}(D)) & \\ \text{---} \swarrow & & \searrow \text{---} \\ (D, J_D^{\max}) & & \text{Max}(D) \end{array}$$

By applying this description of the topology $J_D^{\max}$ in the particular case $D = L(A)$, in conjunction with the fact that for any elements $a, b \in A$, $[P_a] \leq [P_b]$ in $L(A)$ if and only if a power of a is a multiple of b in A (cf. the technical lemma V3.2 [53]), we easily arrive at the following axiomatization of the theory $\mathbb{P}_A^{\max}$ (below we indicate by a list of elements in parentheses the ideal generated by them): the theory $\mathbb{P}_A^{\max}$ is obtained from $\mathbb{P}_A$ by adding all the sequents of the form

$$\left(P_c \vdash \bigvee_{i \in I} P_{c_i} \right)$$

for any elements $\{c_i \mid i \in I\}$ such that for any $i \in I$ a power of c_i is a multiple of c in A and for any finite set of elements $d_1, \dots, d_n \in A$ such that $(c, d_1, \dots, d_n) = A$ there exists a finite subset $J \subseteq I$ such that the ideal generated by the d_i (for $i = 1, \dots, n$) and the c_i (for $i \in J$) is the whole of A .

Notice that the compactness of the space $\text{Max}(A)$ can be expressed in logical terms as the assertion that, for any set $\{\phi_i \mid i \in I\}$ of geometric sentences over the signature of $\mathbb{P}_A$, if for every maximal ideal M of A there is an element $i \in I$ such that $M \models \phi_i$ then there exists a finite subset $J \subseteq I$ such that for every maximal ideal M of A , $M \models \phi_i$ for some $i \in J$.

The following result, which is a corollary of the above arguments, provides an explicit description of the sobrification of $\text{Max}(A)$ as a subspace of $\text{Spec}(A)$.

Theorem 10.7.1 *Let A be a commutative ring with unit. Then a prime ideal P of A belongs to the sobrification of $\text{Max}(A)$ (resp. is maximal, if $\text{Max}(A)$ is sober) if and only if for any elements c and $\{c_i \mid i \in I\}$ of A such that for every $i \in I$ a power of c_i is a multiple of c and for any finite set of elements $d_1, \dots, d_n \in A$ such that $(c, d_1, \dots, d_n) = A$ there exists a finite subset $J \subseteq I$*

such that the ideal generated by the d_j (for $j = 1, \dots, n$) and the c_i (for $i \in J$) is the whole of A , if $c_i \in P$ for all $i \in I$ then $c \in P$. $\square$

Remark 10.7.2 Theorem 10.7.1 notably applies to any commutative C^* -algebra (since its Gelfand spectrum $\text{Max}(A)$ is sober being Hausdorff), giving a characterization of its maximal ideals.

We can ‘functorialize’ Theorem 10.7.1 in a natural way, obtaining a characterization of the ring homomorphisms $f : A \rightarrow B$ such that the induced continuous map $\text{Spec}(B) \rightarrow \text{Spec}(A)$ restricts to a (continuous) map $\text{Max}(B) \rightarrow \text{Max}(A)$.

Let us work once again in the more general context of morphisms of distributive lattices.

Theorem 10.7.3 *Let $f : D \rightarrow D'$ be a morphism of distributive lattices. If*

$$\text{Spec}(f) : \text{Spec}(D') \rightarrow \text{Spec}(D)$$

restricts to a (necessarily continuous map)

$$\text{Max}(D') \rightarrow \text{Max}(D)$$

then f is a morphism of sites

$$(D, J_D^{\max}) \rightarrow (D', J_{D'}^{\max}).$$

The converse holds if $\text{Max}(D)$ is sober.

Proof The thesis arises from the consideration of the following ‘bridge’ in light of the general theory of morphisms of sites:

$$\begin{array}{ccc}
 \text{Sh}(D', J_{D'}^{\max}) & \simeq & \text{Sh}(\text{Max}(D')) \\
 \downarrow & & \downarrow \\
 \text{Sh}(D, J_D^{\max}) & \simeq & \text{Sh}(\text{Max}(D)) \\
 \uparrow & \text{---} & \downarrow \\
 (D, J_D^{\max}) & & \text{Max}(D') \\
 & & \downarrow \\
 & & \text{Max}(D)
 \end{array}$$

If $\text{Spec}(f) : \text{Spec}(D') \rightarrow \text{Spec}(D)$ restricts to a (necessarily continuous map) $\text{Spec}(f)| : \text{Max}(D') \rightarrow \text{Max}(D)$ then the following diagram (where the vertical functors are the canonical ones) commutes:

$$\begin{array}{ccc}
 D & \xrightarrow{f} & D' \\
 \downarrow & & \downarrow \\
 \text{Sh}(\text{Max}(D)) & \xrightarrow{\text{Sh}(\text{Spec}(f)|)^*} & \text{Sh}(\text{Max}(D')).
 \end{array}$$

Composing with the equivalences

$$\mathbf{Sh}(D, J_D^{\max}) \simeq \mathbf{Sh}(\mathrm{Max}(D))$$

and

$$\mathbf{Sh}(D', J_{D'}^{\max}) \simeq \mathbf{Sh}(\mathrm{Max}(D')),$$

we obtain the commutativity of the diagram

$$\begin{array}{ccc} D & \xrightarrow{f} & D' \\ \downarrow & & \downarrow \\ \mathbf{Sh}(D, J_D^{\max}) & \longrightarrow & \mathbf{Sh}(D', J_{D'}^{\max}), \end{array}$$

where the vertical functors are the canonical ones. This in turn implies that f is cover-preserving, whence a morphism of sites $(D, J_D^{\max}) \rightarrow (D', J_{D'}^{\max})$.

Conversely, if f is a morphism of sites $(D, J_D^{\max}) \rightarrow (D', J_{D'}^{\max})$ then it induces a geometric morphism

$$\mathbf{Sh}(f) : \mathbf{Sh}(D', J_{D'}^{\max}) \rightarrow \mathbf{Sh}(D, J_D^{\max}),$$

which corresponds under the Morita equivalences

$$\mathbf{Sh}(D, J_D^{\max}) \simeq \mathbf{Sh}(\mathrm{Max}(D))$$

and

$$\mathbf{Sh}(D', J_{D'}^{\max}) \simeq \mathbf{Sh}(\mathrm{Max}(D'))$$

to a geometric morphism

$$\mathbf{Sh}(\mathrm{Max}(D')) \rightarrow \mathbf{Sh}(\mathrm{Max}(D)).$$

This morphism in turn corresponds, if $\mathrm{Max}(D)$ is sober, to a unique continuous map $\mathrm{Max}(D') \rightarrow \mathrm{Max}(D)$ given by the restriction of $\mathrm{Spec}(f) : \mathrm{Spec}(D') \rightarrow \mathrm{Spec}(D)$ to $\mathrm{Max}(D')$ and $\mathrm{Max}(D)$. $\square$

Applying this result in the context of distributive lattices of the form $L(A)$, we obtain the following

Theorem 10.7.4 *Let $f : A \rightarrow B$ be a homomorphism of commutative rings with unit A and B . If the continuous map $\mathrm{Spec}(f) : \mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$ restricts to a (continuous) map $\mathrm{Max}(B) \rightarrow \mathrm{Max}(A)$ then for any elements c and $\{c_i \mid i \in I\}$ of A such that for every $i \in I$ a power of c_i is a multiple of c and for any finite set of elements $d_1, \dots, d_n \in A$ such that $(c, d_1, \dots, d_n) = A$ there exists a finite subset $J \subseteq I$ such that the ideal generated by the d_j (for $j = 1, \dots, n$) and the c_i (for $i \in J$) is the whole of A , if $(f(c), d'_1, \dots, d'_m) = B$ for some elements $d'_1, \dots, d'_m \in B$ then there exists a finite subset $K \subseteq I$ such that the ideal generated by the d'_k (for $k = 1, \dots, m$) and the $f(c_i)$ (for $i \in K$) is the whole of B . The converse implication holds if $\mathrm{Max}(A)$ is sober. $\square$*

Notice that the above development of a natural analogue, for the maximal spectrum of a commutative ring with unit, of the different points of view available on the Zariski spectrum, has been achieved in a purely canonical way following an implementation of the ‘bridge’ technique; no choices of essentially arbitrary nature have been made.

10.8 Compactness conditions for geometric theories

In this section we shall identify the essential syntactic properties which characterize the geometric theories belonging to a proper fragment of geometric logic such as cartesian, regular or coherent logic.

We shall say that an object A of a Grothendieck topos $\mathcal{E}$ is *stably irreducible* if it is irreducible and, for any arrows $f, g : B \rightarrow A$ in $\mathcal{E}$, where B is an irreducible object, the domain of the pullback of f along g is irreducible.

Given a small site $(\mathcal{C}, J)$, we denote by $l_J^{\mathcal{C}} : \mathcal{C} \rightarrow \mathbf{Sh}(\mathcal{C}, J)$ the composite of the Yoneda embedding $y : \mathcal{C} \rightarrow [\mathcal{C}^{\text{op}}, \mathbf{Set}]$ with the associated sheaf functor $a_J : [\mathcal{C}^{\text{op}}, \mathbf{Set}] \rightarrow \mathbf{Sh}(\mathcal{C}, J)$.

Recall that a small site $(\mathcal{C}, J)$ is said to be *regular* (resp. *coherent*) if $\mathcal{C}$ is cartesian and J is generated by sieves generated by single arrows (resp. by finite families of arrows).

Proposition 10.8.1 *Let $(\mathcal{C}, J)$ be a small site.*

- (i) *If $\mathcal{C}$ is cartesian and J is the trivial topology on it then, for every $c \in \mathcal{C}$, $l_J^{\mathcal{C}}(c)$ is a stably irreducible object of $\mathbf{Sh}(\mathcal{C}, J)$.*
- (ii) *If $(\mathcal{C}, J)$ is regular then, for every $c \in \mathcal{C}$, $l_J^{\mathcal{C}}(c)$ is a supercoherent object of $\mathbf{Sh}(\mathcal{C}, J)$.*
- (iii) *If $(\mathcal{C}, J)$ is coherent then, for every $c \in \mathcal{C}$, $l_J^{\mathcal{C}}(c)$ is a coherent object of $\mathbf{Sh}(\mathcal{C}, J)$.*
- (iv) *If every J -covering sieve on an object c of $\mathcal{C}$ contains a finite family generating a J -covering sieve, then $l_J^{\mathcal{C}}(c)$ is a compact object of $\mathbf{Sh}(\mathcal{C}, J)$.*

Proof A proof of parts (ii) and (iii) can be found in [55] (cf. Theorem D3.3.7 and Remark D3.3.10 therein), while part (iv) was proved in [20] (Proposition 2.6(i) therein). So it remains to prove part (i).

The fact that $l_J^{\mathcal{C}}(c) = y(c)$ is irreducible follows from Proposition 10.3.3. In fact, the category $\mathcal{C}$ being cartesian and hence Cauchy-complete, in the topos $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$ the irreducible objects are precisely, up to isomorphism, those of the form $y(c)$ for some c . The fact that $y(c)$ is stably irreducible thus follows from the fact that the Yoneda embedding y is full and preserves pullbacks. $\square$

Thanks to Proposition 10.8.1, we can characterize the classifying toposes of theories lying in fragments of geometric logic in an invariant way:

Theorem 10.8.2 *Let $\mathcal{E}$ be a Grothendieck topos.*

- (i) *$\mathcal{E}$ is the classifying topos of a cartesian theory if and only if it has a separating set of stably irreducible objects which is closed in $\mathcal{E}$ under finite limits.*

- (ii) $\mathcal{E}$ is the classifying topos of a regular theory if and only if it has a separating set of supercoherent objects which is closed in $\mathcal{E}$ under finite limits.
- (iii) $\mathcal{E}$ is the classifying topos of a coherent theory if and only if it has a separating set of coherent objects which is closed in $\mathcal{E}$ under finite limits.

Proof

(i) The ‘only if’ direction follows at once from Proposition 10.8.1(i). Conversely, suppose that $\mathcal{E}$ has a separating set $\mathcal{C}$ of stably irreducible objects which is closed in $\mathcal{E}$ under finite limits; then $\mathcal{C}$, regarded as a full subcategory of $\mathcal{E}$, is cartesian and the Comparison Lemma (Theorem 1.1.8) yields an equivalence $\mathcal{E} \simeq [\mathcal{C}^{\text{op}}, \mathbf{Set}]$. Our thesis thus follows from Theorems 1.4.13 and 2.1.8.

(ii) and (iii) The ‘only if’ direction follows from Theorem D3.3.7 [55] and Remarks D3.3.9–10 [55]. Conversely, suppose that $\mathcal{E}$ has a separating set $\mathcal{G}$ of coherent (resp. supercoherent) objects which is closed in $\mathcal{E}$ under finite limits. Then $\mathcal{G}$ is a cartesian category and, by the Comparison Lemma (Theorem 1.1.8), $\mathcal{E}$ is equivalent to $\mathbf{Sh}(\mathcal{G}, J_{\mathcal{E}|_{\mathcal{G}}})$; now, the site $(\mathcal{G}, J_{\mathcal{E}|_{\mathcal{G}}})$ is clearly coherent (resp. regular) and hence $\mathcal{E}$ is the classifying topos of a regular (resp. coherent) theory (cf. the proof of Theorem 2.1.11). $\square$

Remark 10.8.3 From the proof of Theorem 10.8.2 it is clear that one can equivalently replace ‘stably irreducible’ (resp. ‘supercoherent’, ‘coherent’) with ‘irreducible’ (resp. ‘supercompact’, ‘compact’) in its statement.

Let us now interpret the notions of compact and supercompact object of a Grothendieck topos in the context of the geometric syntactic site of a geometric theory; together with the concept of irreducible object, these notions will be the key ingredients for establishing our characterizations of the geometric theories which are cartesian, regular or coherent over their signature.

Definition 10.8.4 Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\phi(\vec{x})$ a geometric formula-in-context over Σ .

- (a) We say that $\phi(\vec{x})$ is $\mathbb{T}$ -compact if, for any family $\{\psi_i(\vec{x}) \mid i \in I\}$ of geometric formulae over Σ in a given context $\vec{x}$, $(\phi \vdash_{\vec{x}} \bigvee_{i \in I} \psi_i)$ provable in $\mathbb{T}$ implies $(\phi \vdash_{\vec{x}} \bigvee_{i \in I'} \psi_i)$ provable in $\mathbb{T}$ for some finite subset I' of I .
- (b) We say that $\phi(\vec{x})$ is $\mathbb{T}$ -supercompact if, for any family $\{\psi_i(\vec{x}) \mid i \in I\}$ of geometric formulae over Σ in a given context $\vec{x}$, $(\phi \vdash_{\vec{x}} \bigvee_{i \in I} \psi_i)$ provable in $\mathbb{T}$ implies $(\phi \vdash_{\vec{x}} \psi_i)$ provable in $\mathbb{T}$ for some $i \in I$.

Lemma 10.8.5 (cf. Lemma 3.2 [20]). *Let $\mathbb{T}$ be a geometric theory over a signature Σ , $\phi(\vec{x})$ a geometric formula-in-context over Σ and $y_{\mathbb{T}} : \mathcal{C}_{\mathbb{T}} \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$ the Yoneda embedding. Then*

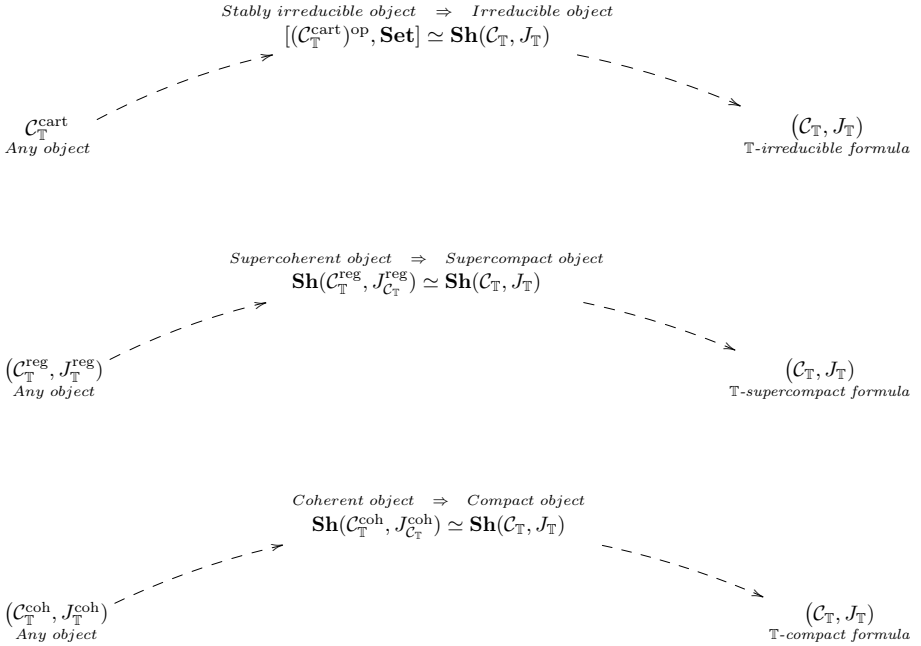
- (i) $\phi(\vec{x})$ is $\mathbb{T}$ -irreducible (in the sense of Definition 6.1.9) if and only if $y_{\mathbb{T}}(\{\vec{x} \cdot \phi\})$ is an irreducible object of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$.
- (ii) $\phi(\vec{x})$ is $\mathbb{T}$ -compact if and only if $y_{\mathbb{T}}(\{\vec{x} \cdot \phi\})$ is a compact object of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$.
- (iii) $\phi(\vec{x})$ is $\mathbb{T}$ -supercompact if and only if $y_{\mathbb{T}}(\{\vec{x} \cdot \phi\})$ is a supercompact object of $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$.

The following theorem, whose part (ii) extends Lemma D3.3.11 [55], is obtained from Proposition 10.8.1 and Lemma 10.8.5 by applying the ‘bridge’ technique (cf. section 2.2).

Theorem 10.8.6 *Let $\mathbb{T}$ be a geometric theory over a signature Σ and $\phi(\vec{x})$ a geometric formula-in-context over Σ .*

- (i) *If $\mathbb{T}$ is cartesian and $\phi(\vec{x})$ is $\mathbb{T}$ -cartesian then $\phi(\vec{x})$ is $\mathbb{T}$ -irreducible.*
- (ii) *If $\mathbb{T}$ is regular and $\phi(\vec{x})$ is regular then $\phi(\vec{x})$ is $\mathbb{T}$ -supercompact.*
- (iii) *If $\mathbb{T}$ is coherent and $\phi(\vec{x})$ is coherent then $\phi(\vec{x})$ is $\mathbb{T}$ -compact.*

Proof The theorem arises from the consideration of the following ‘bridges’:



Specifically, if $\mathbb{T}$ is a cartesian (resp. regular, coherent) theory then for any $\mathbb{T}$ -cartesian (resp. regular, coherent) formula $\phi(\vec{x})$ over Σ we have that $y_{\mathbb{T}}^{\text{cart}}(\{\vec{x} \cdot \phi\})$

(resp. $y_{\mathbb{T}}^{\text{reg}}(\{\vec{x} \cdot \phi\})$, $y_{\mathbb{T}}^{\text{coh}}(\{\vec{x} \cdot \phi\})$) and $y_{\mathbb{T}}(\{\vec{x} \cdot \phi\})$ correspond to each other under the equivalence

$$[(\mathcal{C}_{\mathbb{T}}^{\text{cart}})^{\text{op}}, \mathbf{Set}] \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})$$

(resp.

$$\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, J_{\mathcal{C}_{\mathbb{T}}^{\text{reg}}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}),$$

$$\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, J_{\mathcal{C}_{\mathbb{T}}^{\text{coh}}}) \simeq \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}})),$$

where $y_{\mathbb{T}}^{\text{cart}} : \mathcal{C}_{\mathbb{T}}^{\text{cart}} \hookrightarrow [(\mathcal{C}_{\mathbb{T}}^{\text{cart}})^{\text{op}}, \mathbf{Set}]$ (resp. $y_{\mathbb{T}}^{\text{reg}} : \mathcal{C}_{\mathbb{T}}^{\text{reg}} \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, J_{\mathbb{T}}^{\text{reg}})$, $y_{\mathbb{T}}^{\text{coh}} : \mathcal{C}_{\mathbb{T}}^{\text{coh}} \hookrightarrow \mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{coh}}, J_{\mathbb{T}}^{\text{coh}})$) is the Yoneda embedding.

Now, by Proposition 10.8.1, $y_{\mathbb{T}}^{\text{cart}}(\{\vec{x} \cdot \phi\})$ (resp. $y_{\mathbb{T}}^{\text{reg}}(\{\vec{x} \cdot \phi\})$, $y_{\mathbb{T}}^{\text{coh}}(\{\vec{x} \cdot \phi\})$) is a stably irreducible and hence irreducible (resp. supercoherent and hence supercompact, coherent and hence compact) object of the topos $[(\mathcal{C}_{\mathbb{T}}^{\text{cart}})^{\text{op}}, \mathbf{Set}]$ (resp. $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{reg}}, J_{\mathbb{T}}^{\text{reg}})$, $\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}^{\text{coh}}, J_{\mathbb{T}}^{\text{coh}})$). Our thesis thus follows from Lemma 10.8.5. $\square$

Theorem 10.8.7 (cf. Theorems 3.3-4-5 [20]). *Let $\mathbb{T}$ be a geometric theory over a signature Σ . Then $\mathbb{T}$ is syntactically equivalent to a cartesian (resp. regular, coherent) theory over Σ if and only if every $\mathbb{T}$ -cartesian (resp. regular, coherent) formula over Σ is $\mathbb{T}$ -irreducible (resp. $\mathbb{T}$ -supercompact, $\mathbb{T}$ -compact).*

Proof The ‘only if’ part of the theorem follows from Theorem 10.8.6. To prove the converse direction we observe that, by Lemma D1.3.8 [55], the full subcategory of the geometric syntactic category $\mathcal{C}_{\mathbb{T}}$ of $\mathbb{T}$ on the Horn formulae is $J_{\mathbb{T}}$ -dense. So if every $\mathbb{T}$ -cartesian formula over Σ is $\mathbb{T}$ -irreducible the Comparison Lemma (Theorem 1.1.8) yields an equivalence

$$\mathbf{Sh}(\mathcal{C}_{\mathbb{T}}, J_{\mathbb{T}}) \simeq [(\mathcal{C}_{\mathbb{T}'}^{\text{cart}})^{\text{op}}, \mathbf{Set}],$$

where $\mathbb{T}'$ is the cartesianization of $\mathbb{T}$ (i.e. the cartesian theory consisting of all the $\mathbb{T}$ -cartesian sequents which are provable in $\mathbb{T}$), which shows that $\mathbb{T}$ is syntactically equivalent to $\mathbb{T}'$, as required. Similar arguments work for the regular and coherent cases. $\square$

